

Reverse-engineering of T cell proliferation dynamic

By

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DECLARATION

This work was carried out at AIMS-Ghana in partial fulfilment of the requirements for a Master of Science in Mathematical Sciences.

I hereby declare that except where due acknowledgement is made, this work has never been presented wholly or in part for the award of a degree at AIMS-Ghana or any other University.

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DEDICATION

To my family especially to my parents and Dr. Antoine Tambue

List of abbreviations

ARE	Average relative error
CFDA,SE	Carboxyfluorescein succinimidyl ester (nonfluorescing)
CFSE	Carboxyfluorescein succinimidyl ester
CTL	Cytotoxic T lymphocytes
DDE	Delay differential equations
DNA	Deoxyribonucleic acid
DMSO	Dimethyl sulfoxide
FCS	Fetal calf serum
IBS	Individual based simulation
MHC	Major histocompatibility complex
NK	Natural killer
ODE	Ordinary differential equation
PBS	Phosphate buffered saline
RMPI	Roswell Park Memorial Institute
SSD	Sum of square of differences
TCD4	Helper T-cells
TCD8 or CTL	Killer T-cells

Contents

Declaration	i
Acknowledgements	ii
Dedication	iii
List of abbreviations	iv
Abstract	1
1 Introduction	2
2 Experiment and Theory	4
2.1 Procedures for CFSE experiment	4
2.2 Description of Models	6
3 Results	22
3.1 Simulation	22
3.2 Further simplifications of the Cyton Model	24
3.3 Analysis of Smith-Martin Model and Cyton Model	29
4 Discussion and Conclusion	34
5 Appendix	36
5.1 Codes	36
References	50

Abstract

T cells are of paramount importance for human well-being. This explains why it is necessary to understand parameters that govern T cells' proliferation dynamics. This essay aims to review four main mathematical models used so far to describe T cell proliferation, including the Smith-Martin and Cyton models. A brief description of a labeled carboxyfluorescein succinimide ester (CFSE) experimental protocol, used to measure T cell proliferation, is presented. This allows for understanding the main assumptions used to build the models. In addition, we derive a closed-form expression for the Cyton Model in the case when waiting times for T cells to die and divide are exponentially distributed. Numerical data such as a CFSE experiment could provide are generated with the aid of an individual-based simulation. These serve as the basis for testing and comparing the performances of the Smith-Martin and Cyton models.

1. Introduction

Every day, we are subject to external attacks. Why? Because the human body offers good environments for microbial development. Most of the time, the attacks are stealthy and since we are naturally protected by our *immune system*, we do not know that we are assaulted.

We can define the immune system as a connected and orchestrated set of cells, tissues, and organs that work together to defend the body against external attacks such as microbes (bacteria, virus, fungi, etc.), parasites (worms, tiny, fleas, lice, etc) and other attacks such as small particles of dust or internal attacks such as cancer (see reference [1]). Sometimes, we are sick or weak which is an indication that our immune system is not always victorious in its battles, explaining why it is of paramount importance to understand how it works. This has been the subject of several intensive debates driven theoretically and/or experimentally due to versatile applications in biology, pharmacology, nanotechnology, etc.

The immune system is complex. One of the most comprehensive descriptions involves dividing it into two branches: the innate branch and the adaptive branch; each of which has a specific structure. In order to provide a good description of each, we have to introduce the notion of *self* and *non-self*. The *self* is the set of cells and molecules that belong to the body, and *non-self* is everything that can trigger an immune response that is those which do not carry self markers. When the body is assaulted by a foreign body, the immune system identifies the threat because it does not bear self markers and quickly responds by preparing its "soldiers" to carry out the "invaders". The first reaction is due to the innate branch, consisting consists of cells and proteins, that are always ready to fight the "invaders" regardless of what they are. Among other things, we can mention the main components that are physical epithelial barriers, phagocytic leukocytes, dendritic cells, and a special type of lymphocyte called *natural killer* (NK) cells. When the innate branch has failed to eradicate the pathogens, the second branch (adaptive branch) takes over.

As mentioned, the adaptive immunity takes place when the innate branch fails. It is directed by cells called lymphocyte that are of two types: B and T cells. Before talking about T cells that are the main interest of this work, let us give a breve description of B cells' functionality. B cells are responsible for humoral response. They are composed about of 10% lymphocytes, are always immature, and circulate in the bloodstream. Their function is to integrate antibodies directly into the cell cytoplasm. If we are attacked by a pathogen, B cells complete their maturation and proliferate in order to provide an appropriate immune response by turning into plasma cells, which secrete antibodies whose role is to recognize a specific antigen. These antibodies bind to antigens due to their complementary structure, leading to phagocytosis given that phagocytes only recognize the complex antigen-antibody. At the end of this response, some B cells are stored in the memory of the immune system for a faster and more efficient response to future attacks. The body can also use the cellular response for certain types of invasions like viral invasions. This kind of response is directed by T cells (for further description of B cell's mechanism, see [1]).

T cells are the key to the immune system because they coordinate the function of adaptive immune response. They can be divided into two categories: helper T cells (TCD4), which orchestrate an immune response and play important roles in all arms of immunity by communicating with other cells; and killer T cells or cytotoxic T lymphocytes (CTLs or TCD8) which recognize cells

carrying certain foreign or abnormal molecules on their surfaces. This is why they are very useful against viral nature threats-viruses often hide and grow inside infected cells. Such foreign proteins entering a cell by infection undergo intracellular partial degradation and are transported through the major histocompatibility complex (MHC) proteins on the cell surface to be presented to the cytotoxic T cell. This mechanism kills the host cell directly by inducing a programmed cell death (apoptosis). Some antigens are important for our body, but they do not carry the self makers, they should be eradicated. However, this does not happen because some T cells called regulatory T cells or suppressor T cells regulate the immune response that enable these useful cells to survive. Others types are Natural Killer T cells (NK) and Gamma-delta T cells which are part of both the innate branch and the adaptive branch. As we saw earlier, the adaptive of the immunity system is important for the body, since it is coordinated by T cells. The T cells must proliferate such that their population becomes sufficiently large enough to ensure rapid and efficient response.

In some cases, it is necessary to artificially enhance T-cell proliferation (e.g. using drugs) to enable a more effective response to specific health threats. This requires altering key parameters that control T cell proliferation dynamics. Therefore, it is useful to know what those control parameters are and quantify them. In other words, it is necessary to reverse-engineer T cell proliferation dynamics. Experimental methods such as Carboxy-fluorescein diacetate succinimidyl ester (CFSE) labeling have been used to measure T cell proliferation dynamics. However, these methods produce quantitative readouts that tell us little about the underlying control parameters and their values. We need to use well-formulated mathematical models to infer those parameters from the experimental readouts.

In this essay, we review four mathematical models currently used to analyze the readouts from CFSE experiments in order to quantify parameters that control T-cell proliferation. We elucidate the key assumptions of the models, and identify their strengths and weakness as well as areas where they can be improved. We develop an individual-based simulation of the CFSE experimental process to evaluate how accurately the reviewed mathematical models describe T-cell proliferation as predicted by CFSE experiments. We will first describe the CFSE experimental procedure, the mathematical models, and the simulation algorithm.

2. Experiment and Theory

2.1 Procedures for CFSE experiment

Several experimental techniques have been used to estimate the lymphocyte dynamics. Among other, the 5-bromo-2-deoxyuridine labeling, the deuterium labeling (Brdu) [2], the deuterated glucose (2H_2 -glucose) [3] have been widely used. These methods describe the proliferation kinetics without providing the history of the cell. Importantly, another method that was initiated by Lyon and Parish, the Carboxy-fluorescein diacetate succinimidyl ester (CFSE) labeling, is one of most employed methods used to track and quantify cell dynamics. The advantage of using this method is that it maintains cell viability to a greater degree during the manipulation, and that the maximum number of cell divisions can be observe [4].

CFSE is used to monitor the division of cells such as lymphocytes within several months [4]. The coloration density provides information about the cell's generation, starting with its ancestor. It can also be used to track the lymphocyte's migration and positioning [4]. In order to determine the set of parameters governing T cells' proliferation, it is of paramount importance to describe the process of monitoring T cell proliferation using CFSE labeling. Initially, the CFSE is not fluorescent and, due to the presence of two acetate groups from a compound, this facilitates the permeability of the cell's plasma membrane. Once it enters into the cell, the acetate groups is eliminated and the CFSE becomes fluorescent [4]. Thus, the idea consists of labeling an initial set of long-lived intracellular molecules with the highly fluorescent dye called carboxyfluorescein and following cell divisions. Note that the labeling process typically takes less than an hour to complete. At this initial stage, the dye is equivalently distributed into molecules. As a consequence, the fluorescence of daughter cells (cells descended from this generation by division) is reduced to half to that of their parents. It should also be noted that the daughter cells can be identified up the 8th generation. Thus, the number of resulting cells after each generation can be quantified by flow cytometry, a technique that accounts for the intracellular dye intensity.

Before describing the protocol of monitoring T cells proliferation using CFSE labeling, let us summarize the typical characteristics of the method. The following characteristics are inspired from study made by Quah et al. [4] and Quah and Parish [5]:

- The capacity to adapt to long-lived processes;
- The equal distribution of fluorescent to progenies of the same generation;
- The reduction to half of the fluorescence between two consecutive generations;
- The labeling protocol takes less than one hour to perform;
- Cells are tracked up to a maximum of 8 divisions.

2.1.1 Protocol.

2.1.2 Materials.

Now in order to describe the protocol for monitoring T cells dynamics using the CFSE dye, we refer to the description made by Quah and Parish [5]. We start by presenting the list of materials necessary to fully realize the experiment. Thence, we need: carboxyfluorescein diacetate succinimidyl ester (CFDA, SE) (MW 557.47 *g*/mole) which will be converted into CFSE; dimethyl sulfoxide (DMSO); fresh T cell from human or from mouse; RPMI 1640 medium; fetal calf serum (FCS); 10-15 *mL* of canonical tubes; Phosphate buffered saline (PBS); aluminium foil; vortex, centrifuge and flow cytometer.

2.1.3 Reagent setup.

The reagent setup as described in this subsection is inspired by Quah and Parish [5] on one hand, and Quah et al. [4] on the other hand.

- Dissolve 25mg of CFDA, SE power in 8,96 *mL* of DMSO to obtain final solution of 5*mM*. (be careful because the CFSE reacts rapidly with the solution DMSO. It is advised to do this mixture just before the cell labeling.
- Freshly isolated T cell from spleen or lymph nodes of human or mice and resuspend in 1 *mL* of culture medium (RMPI) with 5% heat-inactivated (HI) fetal calf serum (FCS), with a cell concentration of between 0.5×10^6 - 10×10^7 *mL*.
- For cell purification reagent, one uses negative selection procedure by using magnetic bead technologies, positive or negative selection procedures using magnetic microbeads or cell sorting using Immunofluorescence flow cytometry.
- Use the appropriate stimuli to encourage the proliferation, for instance phytohemagglutinin is indicated for human T cell and ovalbumin for mouse.
- To distinguish the dead cells from viable cells, propidiumiodide or 7-aminoactinomycin D is indicated.

2.1.4 CFSE labeling of T cells.

For the Labeling, we describe the routine method given by [5]:

- Thoroughly resuspend cells in the 1 *mL* volume of culture medium and place carefully in the bottom of a fresh (non-wetted) 10 *mL* conical tube.
- Hold the tube horizontally to prevent premature mixing of suspended cell and CFDA, SE solution.
- Add 110 μ *L* of PBS to the dry part at the top tube and carefully to avoid contact with cell solution.
- Resuspend 1.1 μ *L* of the 5 *mM* stock of CFDA, SE in the 110 μ *L* of PBS.
- Quickly mixing to obtain uniform mixture

- Cover the tube with aluminum foil to protect it from the light and incubate cells for five minutes at room temperature.
- Wash cells by diluting in 10 volumes of 20 degree Celsius of PBS containing 5% of HI FCS, sedimenting by centrifugation at $300\times g$ for 5 min at 20 degree Celsius and discarding the supernatant. Repeat washing twice more.
- Proceed to the counting by using cytometry flow.

The summary of events occurring during CFSE labeling of T cells is described by figure 2.1.

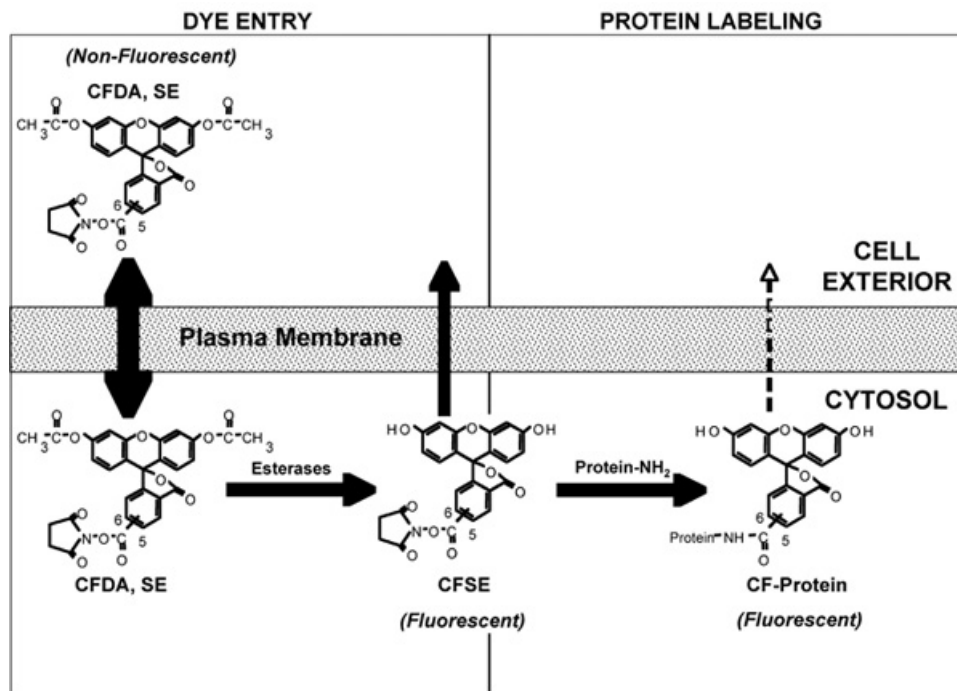


Figure 2.1: A representation of the various molecular events that occur during the labeling of cells with carboxyfluorescein diacetate succinimidyl ester (CFDA, SE) Image from [5]

2.2 Description of Models

It is common in general to move from a problem to a solution. In this work, we rather consider a reverse-engineering procedure. We use existing solution to extract information that will help us to solve the general problem. We simulate data as designed CFSE experiments could provide and attempt to construct mathematical models that reflect as much as possible the faced situation and simultaneously describe T cells' proliferation dynamics. As far as this work is concerned, we focus our attention on four specific models. By order of accuracy as mentioned by Miao et al. [6], we start by considering a Smith-Martin-like model (Smith and Martin [7]), later turn-on to a linear birth-death ordinary differential equation (ODE) model for slowly diving cells and finally end up with the cyton model and the age-dependent multitype branching process.

2.2.1 Smith-Martin Model and Linear Birth-death Model.

The *Smith-Martin Model*, named after the authors J. A. Smith and L. Martin, was originally formulated to mathematically answer the key question of cell cycles. It was introduced in 1973 and has become the plinth of numerous and remarkable studies in mathematical modeling of cells proliferation. The model in its original form, has undergone several modifications that allow it within some circumstances to fit the data of some experiments. In the present piece of work, we discuss a version of the Smith-Martin Model widely used in order to interpret CFSE data of T cells dynamics ([8], [9], [10], [11], [12]). Such models can be written as set coupled delayed differential equation (DDE) that describe the rate of population change over the time and mimics ecological predator prey models [9].

As it's well known, a cell's cycle consists of four phases: G_1 (interval between mitosis and DNA synthesis), S (period of DNA synthesis), G_2 (interval between S and mitosis) and M (mitosis) (see figure 2.2). Smith and Martin have proposed to divide these phases into two fundamental phases: the A phase and the B phase. In the A phase, the activity of cells is non-deterministic. A cell's activity may not be directed towards replication or the cell can remain therein for any length of time. In this last situation, the cell probability of leaving the A phase in order to enter the B phase is constant. This suggests that the distribution of time for the first division obeys a normal law. Being in the B phase, their activities are deterministic. Below are listed a few assumptions relevant the Smith-Martin-like model. Later on, we present the model and highlight some relevant limitations.

- (i) The cell's life is a perpetual cycle that is divided into four phases: the mitotic phase (M) and the inter-mitotic stage compound of DNA synthesis (S), the period between the mitosis and DNA synthesis (G_1), and the period between the DNA synthesis and mitosis (G_2).
- (ii) The A phase contains part or all of the G_1 phase. The B phase is composed of S , G_2 , M , and sometimes part of the G_1 phase, since the G_1 phase is not fixed. The B phase is generally considered to be fixed. The length of the G_1 phase is discussed in [7]. These observations are summarized and illustrated by the figure (2.2).

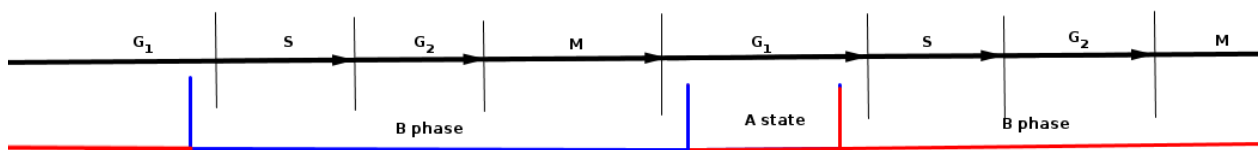


Figure 2.2: Description of cells cycle on the assumptions of Smith-Martin Model

This description is mainly theoretical because Smith and Martin [7] have shown that the initiation of the cell's replication occurs randomly, therefore is not really a cycle. The selection of cells from A phase to B phase is random. Once in B phase, the cells stay for a fixed time Δ (length of the B phase) and divide at the end of the B phase given that mitosis (M) belongs to the B phase. The daughter cells resulting from the division directly enters the A phase. It is also important to note that the number of cells that exit the B phase per unit of time is equal to the number of cells that enter per unit of time : $\lambda N_{n-1}^A(t - \Delta)$, $n = 1, 2, 3 \dots$. It is assumed that A phase

and B phase have the same death rate d_0 for 0 division and d for n divisions ($n = 1, 2, 3, \dots$). The rate of activation (from A phase to B phase) is λ_0 for 0 division and λ for n divisions, $n = 1, 2, 3, \dots$ [8]. Experiments have shown that when the quiescent cells (0 division) are activated to proliferate, the B phase has a length $\Delta_0 > \Delta$ and the rate of activation $\lambda_0 < \lambda$. Thus, for the version of Smith-Martin Model as described here, $\Delta_0 = \Delta$. This assumption is considered by Miao et al. [6].

Let $N_n^A(t)$ and $N_n^B(t)$ respectively represent the number of cells in the A phase and B phase after n divisions ($n=0,1,2,\dots$).

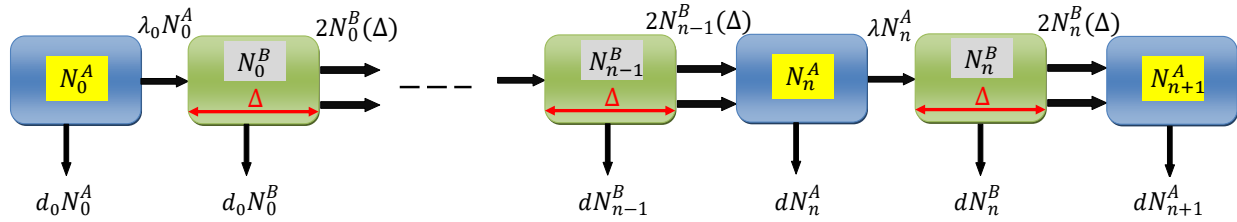


Figure 2.3: Compartment description of cells cycle

The number of cells in the A phase at $t + \delta t$ which have undergone 0 division is equal to number of cell present at time t minus the fraction of cells which leaves the A phase to the B phase minus the fraction of cells which die. So, we have:

$$N_0^A(t + \delta t) = N_0^A(t) - (\lambda_0 + d_0)N_0^A(t)\delta t, \quad (2.2.1)$$

and the number of cells in the B phase at $t + \delta t$ which have undergone 0 division is equal to the number of cells present at time t minus the the fraction of cells alive which leaves the B phase at $t - \Delta$ to A phase (Since after a time Δ , cell must leave B phase. Note also that $e^{-d_0\Delta}$ is the probability that a cell is alive Δ units after entering the B phase), minus the fraction of cells which died. Then, we have

$$N_0^B(t + \delta t) = N_0^B(t) + [\lambda_0 N_0^A(t) - \lambda_0 e^{-d_0\Delta} N_0^A(t - \Delta) - d_0 N_0^B(t)]\delta t. \quad (2.2.2)$$

The number of cells in the A phase after one division at time $t + \delta t$ is equal to number the number of cells present at time t plus twice the number of cells from B phase (since after the B phase cells divide) minus the fraction of cells which leaves from A phase to B phase. So, we have

$$N_1^A(t + \delta t) = N_0^B(t) + [2\lambda_0 e^{-d_0\Delta} N_0^A(t - \Delta) - (\lambda + d)N_1^A(t)]\delta t. \quad (2.2.3)$$

Using the same reasoning, we obtain the number of cells present in the A phase and in B phase at time $t + \delta t$ after realizing n divisions:

$$N_n^A(t + \delta t) = N_n^A(t) + [2\lambda e^{-d\Delta} N_{n-1}^B(t - \Delta) - (\lambda + d)N_n^A(t)]\delta t. \quad n \geq 2 \quad (2.2.4)$$

$$N_n^B(t + \delta t) = N_n^B(t) + [\lambda N_n^A(t) - \lambda e^{-d\Delta} N_n^A(t - \Delta) - dN_n^B(t)]\delta t. \quad n \geq 1. \quad (2.2.5)$$

We know that if f is a given function of time, its derivative with respect to time reads

$$\frac{df(t)}{dt} = \dot{f}(t) = \lim_{\delta t \rightarrow 0} \frac{f(t + \delta t) - f(t)}{\delta t}. \quad (2.2.6)$$

By applying this idea to equations (2.2.1)-(2.2.5), we obtain

$$\begin{cases} \dot{N}_0^A(t) = -(\lambda_0 + d_0)N_0^A(t) \\ \dot{N}_0^B(t) = \lambda_0 N_0^A(t) - \lambda_0 e^{-d_0 \Delta} N_0^A(t - \Delta) - d_0 N_0^B(t) \\ \dot{N}_1^A(t) = 2\lambda_0 e^{-d_0 \Delta} N_0^A(t - \Delta) - (\lambda + d)N_1^A(t) \\ \dot{N}_n^A(t) = 2\lambda e^{-d\Delta} N_{n-1}^A(t - \Delta) - (\lambda + d)N_n^A(t), \quad n \geq 2 \\ \dot{N}_n^B(t) = \lambda N_n^A(t) - \lambda e^{-d\Delta} N_n^A(t - \Delta) - dN_n^B(t), \quad n \geq 1. \end{cases} \quad (2.2.7)$$

It is worth mentioning that the first of these equations decouples from the other and suggests an exponential decay.

$$N_0^A(t) = N_0^A(0)e^{-(\lambda_0 + d_0)t}, \quad (2.2.8)$$

where $N_0^A(0)$ is the number of cells in A phase at time $t = 0$.

Given that we are interested in the total number of cells at arbitrary time t , and that we cannot distinguish between the A phase and the B -phase (which is one of the main issues with this model in practice), it is adequate to reduce the number of independent variables by defining the auxiliary variable

$$N_n(t) = N_n^A(t) + N_n^B(t), \quad n = 0, 1, 2, 3, \dots \quad (2.2.9)$$

which stands for the number of cells after n division. In order to obtain a simplified form of (2.2.7) which is easy to handle, let us note from (2.2.9) that by taking the time derivative of both sides we obtain:

For $n = 0$,

$$\begin{aligned} \dot{N}_0(t) &= \dot{N}_0^A(t) + \dot{N}_0^B(t) \\ &= -(\lambda_0 + d_0)N_0^A(t) + \lambda_0 N_0^A(t) - \lambda_0 e^{-d_0 \Delta} N_0^A(t - \Delta) - d_0 N_0^B(t) \\ &= -\lambda_0 e^{-d_0 \Delta} N_0^A(t - \Delta) - d_0 (N_0^A(t) + N_0^B(t)) \\ &= -\lambda_0 e^{-d_0 \Delta} N_0^A(t - \Delta) - d_0 N_0(t), \end{aligned} \quad (2.2.10)$$

for $n = 1$,

$$\begin{aligned} \dot{N}_1(t) &= \dot{N}_1^A(t) + \dot{N}_1^B(t) \\ &= 2\lambda_0 e^{-d_0 \Delta} N_0^A(t - \Delta) - (\lambda + d)N_1^A(t) + \lambda N_1^A(t) - \lambda e^{-d\Delta} N_1^A(t - \Delta) - dN_1^B(t) \\ &= 2\lambda_0 e^{-d_0 \Delta} N_0^A(t - \Delta) - \lambda e^{-d\Delta} N_1^A(t - \Delta) - d(N_1^A(t) + N_1^B(t)) \\ &= 2\lambda_0 e^{-d_0 \Delta} N_0^A(t - \Delta) - \lambda e^{-d\Delta} N_1^A(t - \Delta) - dN_1(t) \end{aligned} \quad (2.2.11)$$

and for $n \geq 2$,

$$\begin{aligned}
 \dot{N}_n(t) &= \dot{N}_n^A(t) + \dot{N}_n^B(t) \\
 &= 2\lambda e^{-d\Delta} N_{n-1}^A(t - \Delta) - (\lambda + d)N_n^A(t) + \lambda N_n^A(t) - \lambda e^{-d\Delta} N_n^A(t - \Delta) - dN_n^B(t) \\
 &= 2\lambda e^{-d\Delta} N_{n-1}^A(t - \Delta) - \lambda e^{-d\Delta} N_n^A(t - \Delta) - d(N_n^A(t) + N_n^B(t)) \\
 &= 2\lambda e^{-d\Delta} N_{n-1}^A(t - \Delta) - \lambda e^{-d\Delta} N_n^A(t - \Delta) - dN_n(t).
 \end{aligned} \tag{2.2.12}$$

By considering the result from (2.2.7), (2.2.10), (2.2.11) and (2.2.12) we obtain at this stage:

$$\left\{ \begin{aligned} \dot{N}_0^A(t) &= -(\lambda_0 + d_0)N_0^A(t) \\ \dot{N}_0(t) &= -\lambda_0 e^{-d_0\Delta} N_0^A(t - \Delta) - d_0 N_0(t) \\ \dot{N}_1^A(t) &= 2\lambda_0 e^{-d_0\Delta} N_0^A(t - \Delta) - (\lambda + d)N_1^A(t) \\ \dot{N}_1(t) &= 2\lambda_0 e^{-d_0\Delta} N_0^A(t - \Delta) - \lambda e^{-d\Delta} N_1^A(t - \Delta) - dN_1(t) \\ \dot{N}_n^A(t) &= 2\lambda e^{-d\Delta} N_{n-1}^A(t - \Delta) - (\lambda + d)N_n^A(t), \quad n \geq 2 \\ \dot{N}_n(t) &= 2\lambda e^{-d\Delta} N_{n-1}^A(t - \Delta) - \lambda e^{-d\Delta} N_n^A(t - \Delta) - dN_n(t), \quad n \geq 2. \end{aligned} \right. \tag{2.2.13}$$

The main advantage of this model is that it is written in differential form which easily to handle. However, for the reverse-engineering procedure adopted in this work, the data collected for CFSE labeling may not be enough. It will then be difficult to distinguish between A phase and B phase. This remark also holds for their duration of the two phase that are not also distinguishable from each other. On the other hand, DNA-binding dye experiments offer the possibility to overcome this difficulty, allowing one to clearly distinguish these two phases. The Smith-Martin Model as described here with the CFSE dye is found to be inappropriate to distinguish between both phases. Therefore, the model can be simplified for slowly dividing cells to extent that the duration of the A -phase is exceedingly small compared with the duration of a full cycle [8] and acquires the Linear Birth-death ODE form:

$$\left\{ \begin{aligned} \dot{N}_0(t) &= -(\lambda_0 + d_0)N_0(t) \\ \dot{N}_1(t) &= 2\lambda_0 N_0(t) - (\lambda + d)N_1(t) \\ \dot{N}_n(t) &= 2\lambda N_{n-1}(t) - (\lambda + d)N_n(t), \quad n \geq 2. \end{aligned} \right. \tag{2.2.14}$$

The assumption of slowly dividing cells used to build up this equation is quite relevant given that the first division takes a very long time (65 hours for a loss rate of $0.042h^{-1}$ comparable with the experimental data $0.05h^{-1}$) as compared with the total duration of the process [8]. It

worth nothing that though equation (2.2.14) is solved numerically, for $\lambda_0 = \lambda$, $d_0 = d$ and exact analytic result is found in study made by De Boer and Perelson [8] and reads:

$$\begin{cases} N_0(t) = e^{-(\lambda_0+d_0)t} N(0) \\ N_n(t) = \frac{(2\lambda t)^n}{n!} N_0(t). \end{cases}$$

The Smith-Martin Model performances can be considerably improved but unfortunately, the model remains inappropriate to determine the time to death. This explains in part why new models have been required. Among other things, we should mention the Cyton Model and Age-Dependent Multitype Branching Process as presented and discussed below.

2.2.2 Cyton Model.

The mathematical model for T cells proliferation known today as *Cyton Model* is a consequence of over ten years of reverse-engineering between successive experiments and mathematical modeling by its pioneers and Gett and Hodgkin [13] and Hawkins et al. [14]. The Cyton Model can also be viewed as one of the successful scientific achievements in which experiment and theory are conducted in tandem. The Cyton Model was introduced at the beginning of this century in the precursor works of Gett and Hodgkin [13] who first proposed a mathematical model for CFSE data analysis. The proposed model is simple, easy to implement, useful for comparing between data sets and assume a normal distribution of time for the first division. This simple model by Gett and Hodgkin [13] has attracted tremendous attention and several drastic drawbacks have been detected. For instance, Hyrien and Zand [15] observed some years later that the model is quantitatively and even qualitatively inappropriate to describe a simple branching process. This default was attributed to the fact that the time to the first division is modeled as a normal distribution. In the mean time, the model by Gett and Hodgkin [13] was being revisited by Hawkins et al. [14] in order to include realistic facts. The formal authors assumed that:

- (i) death and division of cells are random/stochastic processes each associated with a specific continuous distribution (probability density function);
- (ii) both processes are statistically independent but compete to determine the fate of the cell;
- (iii) division events occurring one after another reset the mechanism responsible for these processes;
- (iv) only a given fraction of cells after each division are capable to duplicate into the next generation;
- (v) the mechanisms are division dependent (see [16] for ample discussion).

Based on the assumptions (i)-(v), Hawkins et al. [14] came out with the so-called *Cyton Model* which is so far the subject of intensive debate.

Assumptions (i)-(v) are based on experimental observations. Indeed, (i) is adopted after the observation was made, by measuring the survival curve of a system, that individual cell deaths

are not deterministic (randomly occurrences) and that the lifetime of cells describes a continuous distribution.

As an advantage if the Smith-Martin Model discussed in the previous section is that the Cyton Model is robust against the distribution assumption, while the Smith-Martin Model is sensitive to the distribution assumption. The Cyton Model is capable to fit experimental data that Smith-Martin Model might not.

After recalling the historical finger prints of the Cyton Model in mathematical modeling and cell immunology biology, let us introduce the model in its analytic form. To this end, we fix our target to a brief summary description of the model by using the construction from Hawkins et al. [14], not necessary based onto the details of the mathematical modeling. We review the Cyton Model as proposed by Miao et al. [6]. Let us denote by $n = 1, 2, 3, \dots$ the number of division that cells have undergone. The time to divide in the n th divisions is characterized by the continuous distribution function $\phi_n(t)$, and the time of death is characterized by the continuous distribution function $\psi_n(t)$. Thus, the probability for a cell to survive at time t when it enters the n th division is

$$1 - \int_0^t \psi_n(\tau) d\tau. \quad (2.2.15)$$

Similarly, the probability that no cell division occur at time t , given that cells are entering the n th division is

$$1 - \int_0^t \phi_n(\tau) d\tau. \quad (2.2.16)$$

These definitions of probabilities are straightforward and apply to a homogeneous cell population where, for instance, if cells do not divide, they will die. This does not match the reality. Indeed, not all cells in the population can divide. If they avoid death, some may be indifferent to external stimuli [14]. For instance, Pereira et al. [17] show that cells can be non-responsive to stimulation due to the absence of effective receptors. From these remarks, Hawkins et al. [14] introduce the progressor fraction f_n , $n = 0, 1, 2, \dots$, which represents the probability that a cell will divide in response to stimulation. It also accounts for the possibility that some cells will stop dividing at a particular generation and remain there until they die. So, the probability for a cell to survive at time t when it enters the n th division becomes

$$f_n \cdot \left(1 - \int_0^t \psi_n(\tau) d\tau \right), \quad (2.2.17)$$

and the probability that no cell division occurs at time t as cells are entering the n th division becomes

$$1 - f_n \cdot \int_0^t \phi_n(\tau) d\tau. \quad (2.2.18)$$

Let $N(0)$ be the numbers of cells of age zero at time $t = 0$. The rate of cell division in the first stage at time t is equal to $N(0)$ times the probability of not dying up to time t times the

probability of dividing at time t ($f_0 \cdot \phi_0$) times the total number of cells.

$$r_0^{div}(t) = f_0 \cdot N(0) \cdot \left(1 - \int_0^t \psi_0(\tau) d\tau\right) \cdot \phi_0(t), \quad (2.2.19)$$

and the rate of cell death in the first stage at time t is equal to $N(0)$ times the probability for not being divided in the zero step, times the probability of dying after age stage zero

$$r_0^{die}(t) = N(0) \cdot \left(1 - f_0 \int_0^t \phi_0(\tau) d\tau\right) \cdot \psi_0(t). \quad (2.2.20)$$

Given that after each division the rate of cells is doubled, the rate of cells entering the n th division that are subjected to divide is given by $2r_{n-1}^{div}$. Accordingly, parts of a corresponding population divide or die with a certain probability, i.e. one may consider cells entering the n th division at time t' as a set of cells split into two parts: one that divides and another that dies. Therefore, the rate of cells entering the n th division within an infinitesimal time interval $d\tau$ is equivalent to the rate $2r_{n-1}^{div}$ of cells entering the $(n-1)$ division multiplied by the probability that cells survive after the $(n-1)$ division which last $t-\tau$, multiplied by the probability of dividing at time $t-\tau$. Thus, the total rate r_n^{div} at time t corresponds to the integral over intermediate rates

$$r_n^{div}(t) = 2f_n \cdot \int_0^t r_{n-1}^{div}(\tau) \cdot \left(1 - \int_0^{t-\tau} \psi_n(\zeta) d\zeta\right) \cdot \phi_n(t-\tau) d\tau. \quad (2.2.21)$$

Similarly, for dying cells, one has

$$r_n^{die}(t) = 2 \cdot \int_0^t r_{n-1}^{div}(\tau) \cdot \left(1 - f_n \int_0^{t-\tau} \phi_n(\zeta) d\zeta\right) \cdot \psi_n(t-\tau) d\tau. \quad (2.2.22)$$

The progressor fraction f_n is introduced in (2.2.21) and (2.2.22) to account for stimulus. We are now in the position to evaluate the number of cells $N_n(t)$ in each generation as a function of time. The desired quantity is obtained by considering the number of cells that have entered the n generation by division as well as those that have left by means of division and death. Thus,

$$\begin{cases} N_0(t) = N(0) - \int_0^t (r_0^{div}(\tau) + r_0^{die}(\tau)) d\tau \\ N_n(t) = \int_0^t (2r_{n-1}^{div}(\tau) - r_n^{div}(\tau) - r_n^{die}(\tau)) d\tau, \quad n = 1, 2, 3, \dots \end{cases} \quad (2.2.23)$$

Although the Cyton Model is formulated with stochastic considerations on the process of division and death, this formulation is subject to some constraints given that its bases do not always fit experimental results. For instance, Hyrien et al. [18] highlight the fact that the probability of division is completely specified by the distribution of the time to division and that of the time to death. Therefore, it will be relevant to consider a more robust model.

2.2.3 Age-Dependent Multitype Branching Process.

The branching process has proven to be a powerful mathematical tool in studies of cellular phenomena, such as cells proliferation and differentiation, which is a vital process for life. Numerous

authors have developed the theory for this field: Harris [19], Mode [20], Athreya and Ney [21], Jagers [22], Kimmel and Axelrod [23], Hyrien and Zand [15], Yakovlev and Yanev [24], Hyrien et al. [18], Yakovlev et al. [25]. Among these authors, some have used branching process to interpret the CFSE labeling, such as Yates et al. [26], Nordon et al. [27], Hyrien et al. [18]. However, some of the proposed models do not take into account certain important considerations, which has triggered some criticisms. For example Hyrien and Zand [15] proposed a branching process model to explain CFSE data. By comparing the quality of parameter obtained after simulation, they were able to demonstrate that the branching model is better than not only the ODE (ordinary differential equation) linear birth-death model, but also Smith-Martin type models. However, the model was criticized by Miao et al. [6], who pointed out that it does not take into account cell death like Cyton Model does, and does not distinguish the subtype of cells. It is worth mentioning that some parameters that govern cell division, such as the time to divide distribution estimated from the model can be biased. This specifically occurs when the number of dead cells is large.

Macken and Perelson [28] and Kimmel and Axelrod [29] have proposed a Multitype Galton-Watson process to explain the dynamics of cell populations. The main point of the proposed model is associated with the discrete character of the process described. Thus, the given model does not meet the natural criteria of CFSE labeling which is a time-continuous processes. So, Miao et al. [6] proposed a more realistic model which takes into consideration the dead cells and the quiescent cells to reverse the engineering of T cell proliferation from CFSE labeling. Here, they consider a progenitor cell (that which has experienced zero divisions) which can: divides, remain quiescent, or die by apoptosis. The descendants resulting from a division are subject to the same fate, that is to say that each dies, divides, or remains quiescent. Since the goal of the model is the study whether the cell will divide, die, or rest during the process, knowing the current status of the cell, or what happens to the cell before its eventual division or death is not important. Instead, Miao et al. [6] determine the type of the cell depending on the events that happen after a period corresponding to the cell cycle:

- Type 1: Cells that will die. These are cells that will not exist after its lifetime
- Type 2: Cells that will stay quiescent. These are cells that survive throughout the experiment and do not contribute during the process;
- Type 3: Cells that will divide into two independent daughters cells in the end of the cell cycle.

For more explanation of the model, the previous authors underlined some very important points that should be taken into consideration. The first is that the quiescent cell does not represent a stable phenotype given that they are all about to die. They discussed the influence of this type of cell on the formulation of the model. The second point is that the length of a cell cycle is not imposed to the experiment and the model proposed is time-continuous. The model used is an Age-Dependent Multitype Branching Process which in general case is a non-Markovian process, unless the distribution of cell lifetime follows the exponential distribution as mentioned in [6]. It has been introduced by Harris [19] and is today recognized as the Bellman-Hariss process. The model and its extensions have been used by several authors, Crump and Mode [30], Jacobson

[31], Mitov and Yanev [32], Hyrien et al. [33]. These authors introduced the Bellman-Harris like model as a age-dependent 3 types branching process under the following assumptions:

1. The experiments starts at time $t = 0$ with respectively $N^{(1)}(0)$ cells of type 1, $N^{(2)}(0)$ cells of type 2, and $N^{(3)}(0)$ cells of type 3;
2. Cells of any type from any generation evolve independently from others;
3. At the end of a cell lifetime, dying cells are eliminated from the system.
4. The lifetime of quiescent cells is beyond the length of the experiment.
5. Cells that are about to divide give two daughter cells of any of three types at the end of its lifetime. It is important to note that the two daughter cells can be of different types.
6. The lifetime of cells of the same type from the same generation follow the same distribution with the same distribution parameters.
7. The lifetime of cells of the same type but different generations may follow the same distribution with different distribution parameters.

Under these assumptions, a key question about the fate of sister cells issued from the division of a type 3 cell may arise. One may think that since they are coming from the same mother cell after division, their fate should be strongly linked (positively correlated). A similar question has been addressed and discussed by Crump and Mode [30]. On the other hand, the data from this experiment have not presented any significant correlation. Therefore, it will be superfluous to incorporate this dependence into the model. An illustration of the model is depicted in figure 2.4. We have started with a single progenitor of age zero, keeping in mind that it can be type 1 with probability P_1 , type 2 with probability P_2 , and type 3 with $P_3 = 1 - P_1 - P_2$. At the end of its lifetime, if it type 3, it gives two daughters cells of age zero which have the same probabilities of being type 1, type 2, or type 3 (i.e. P_1 , P_2 or P_3). The lifetime distribution X_i of cells that are about to proliferate is represented by the cumulative distribution function (c.d.f) $G_i(t) = Pr(X_i \leq t)$ and the lifetime distribution T_i of cells that are going to die is represented by the c.d.f $D_i(t) = Pr(T_i \leq t)$ where the index i denotes the generation that the cell belongs to.

Yakovlev et al. [25] have pointed out that any time-dependent process (with a given offspring distribution) embedded into a d -type Bellman-Harris branching process may be considered as a d -type Galton-Watson branching process with the same offspring distribution. Miao et al/ subsequently adopted that idea and considered a d -type Galton-Watson process embedded into a Bellman-Harris branching process. Let us recall that the non-Markovian d -type Bellman-Harris process has a continuous distribution of a cell lifetime and the d -type Galton-Watson process which is Markovian has a discrete distribution of cell's lifetime. This suggests that if one embeds a Galton-Watson process into a d -type Bellman-Harris branching process, the discrete correspondence yields a d -type Galton-Watson process with the same offspring distribution represented by the random vector [25]

$$Y_n = (Y_n^{(1)}, Y_n^{(2)}, Y_n^{(3)}, \dots, Y_n^{(d)}). \quad (2.2.24)$$

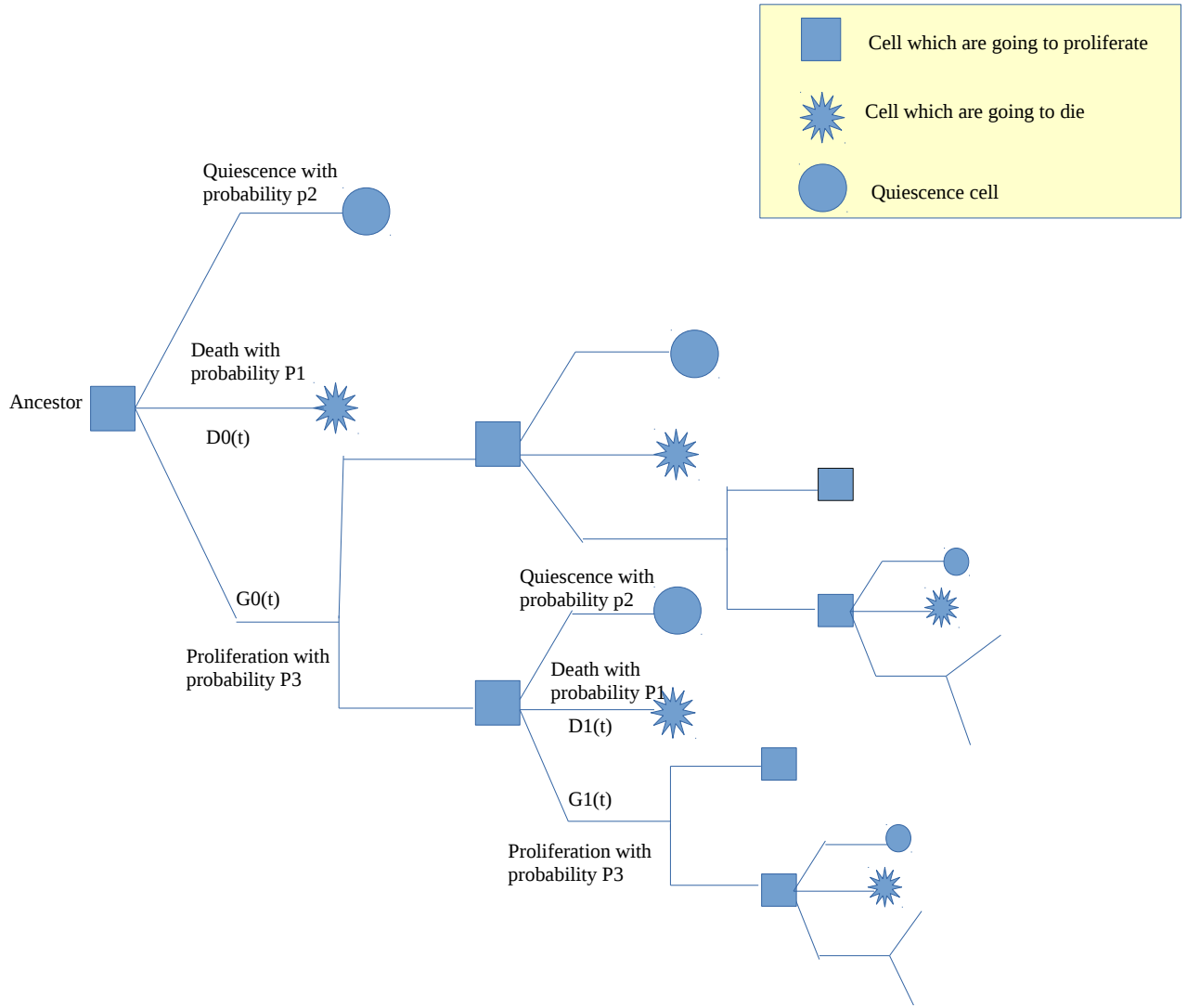


Figure 2.4: Description of branching process inspired from [6]

Let $\xi_j^{(i)}(k, n)$ be the number of cells of type j ($j \in \{1, 2, 3\}$) present in the $(n + 1)$ th generation and derived from the k th cell of type i in the n th generation, and $Y_n^{(j)}$ be the number of cells of type j both dead or alive which were or currently are in the n th generation (that is so to say, cells of type 3 in the n th generation that already divided are also counted although their daughter cells are now in the $(n + 1)$ th generation).

We define the number of cells of types i descending from the cells of type j in n th generation by

$$\mathcal{A}_i^{(j)}(n) := \sum_{k=1}^{Y_n^{(j)}} \xi_i^{(j)}(k, n), \quad n = 0, 1, 2, \dots \quad (2.2.25)$$

Knowing the n th generation, one can compute the $(n + 1)$ th generation by using the following

recurrence relation:

$$Y_{n+1}^{(j)} = \sum_{i=1}^d \mathcal{A}_i^{(j)}(n), \quad j = 1, 2, \dots, d. \quad (2.2.26)$$

For instance, let us consider the system with 3 types of cells ($d = 3$) and let us determine the number of cells of types 3 in the fourth generation. This yields

$$Y_4^{(3)} = \sum_{i=1}^3 \mathcal{A}_i^{(3)}(3), \quad (2.2.27)$$

and

$$\mathcal{A}_i^{(3)}(n) = \sum_{k=1}^{Y_3^{(i)}} \xi_i^{(3)}(k, 3). \quad (2.2.28)$$

Therefore,

$$Y_4^{(3)} = \sum_{k=1}^{Y_3^{(1)}} \xi_1^{(3)}(k, 3) + \sum_{k=1}^{Y_3^{(2)}} \xi_2^{(3)}(k, 3) + \sum_{k=1}^{Y_3^{(3)}} \xi_3^{(3)}(k, 3). \quad (2.2.29)$$

Now, in order to have the number $Z_n^{(j)}(t)$ of cells of type j alive only in a given generation, let us introduce the definition by Athreya and Ney [21] and used by Miao et al. [6] as follows: for

$$k \in \{1, 2, \dots, Y_m^{(j)}\}$$

$$\delta_{n,k}^{(j)}(t) = \begin{cases} 1, & \text{if the } k\text{th cell of type } j \text{ in the } n\text{th generation is alive} \\ & \text{and remains in the } n\text{th generation at time } t \text{ (neither} \\ & \text{died nor finished division);} \\ 0, & \text{otherwise.} \end{cases} \quad (2.2.30)$$

Then,

$$Z_n^{(j)}(t) = \sum_{k=1}^{Y_n^{(j)}} \delta_{n,k}^{(j)}(t). \quad (2.2.31)$$

This formula tells us that once $Y_n^{(j)}$ is known, $Z_n^{(j)}(t)$ is achieved by simply deducting the number of died cells. Remarkably, the formula (2.2.31) is a continuous function of time. As a direct consequence, we moved from a discrete picture to its corresponding continuous counterpart (from Multitype Galton-Watson process embedded to Bellman-Hariss process to Multitype Bellman-Hariss process)

Now, the authors in [6] consider a single cell of type 3, which is a progenitor and compute the conditional mean of the living cells to the n th generation knowing the number of cells alive or died at the n th generation ($E(Z_n^{(3)}(t)|Y_n^{(3)})$). By using the argument from Athreya and Ney [21]

which is to observe the symmetry among the particles and considering the formula (2.2.31), we obtain:

$$\begin{aligned} E(Z_n^{(3)}(t)|Y_n^{(3)}) &= Y_n^{(3)} \cdot E(\delta_{n,1}^{(3)}(t)) \\ &= Y_n^{(3)} \cdot Pr(\delta_{n,1}^{(3)}(t) = 1), \end{aligned} \quad (2.2.32)$$

where $Pr(\delta_{n,1}^{(3)}(t) = 1)$ indicates the probability that a cell of type 3 is alive in the n th generation and remains alive in the n th generation at time t . The authors in [6] have given the condition of having $Pr(\delta_{n,1}^{(3)}(t) = 1)$, namely « a cell of type 3 in the n th generation still and remains alive in the n th generation at time t only when the sum of lifetimes of its ancestors is less or equal to t and the sum of lifetimes of its ancestors and its own is greater than t ». This property is derived from [21].

In order to explicitly get access to $Pr(\delta_{n,1}^{(3)}(t) = 1)$, let us denote by T_h the lifetime of cell of type 3 in the h th generation and consider the statement into the braces «...» above as an event **C**. Therefore, **C** is split as an intersection of two independent events **A** and **B**

$$\mathbf{C} = \mathbf{A} \cap \mathbf{B}. \quad (2.2.33)$$

Here, **A** and **B** are respectively the events «the sum of lifetimes of ancestors of cell of type 3 is less or equal to t » and « the sum of lifetimes of ancestors of cell of type 3 and its own lifetime is greater than t »

$$\mathbf{A} = \sum_{h=0}^{n-1} T_h^{(3)} \leq t, \quad \mathbf{B} = \sum_{h=0}^n T_h^{(3)} > t. \quad (2.2.34)$$

Thus, using the Morgan law of set theory and the fact that the complementary events $\mathbf{A}^c$ and $\mathbf{B}^c$ are mutually excluded, one successively writes

$$\begin{aligned} Pr(\delta_{n,1}^{(3)}(t) = 1) &= Pr(\mathbf{A} \cap \mathbf{B}) \\ &= Pr[(\mathbf{A}^c \cup \mathbf{B}^c)^c] \\ &= 1 - Pr(\mathbf{A}^c \cup \mathbf{B}^c) \\ &= 1 - [Pr(\mathbf{A}^c) + Pr(\mathbf{B}^c)] \\ &= 1 - [1 - Pr(\mathbf{A}) + Pr(\mathbf{B}^c)] \\ &= Pr(\mathbf{A}) - Pr(\mathbf{B}^c) \\ &= Pr\left(\sum_{h=0}^{n-1} T_h^{(3)} \leq t\right) - Pr\left(\sum_{h=0}^n T_h^{(3)} \leq t\right). \end{aligned} \quad (2.2.35)$$

From the properties of Galton-Watson process, we have

$$\begin{cases} Pr\left(\sum_{h=0}^{n-1} T_h^{(3)} \leq t\right) = G_0(t) * G_1(t) * \cdots * G_{n-1}(t) \\ Pr\left(\sum_{h=0}^n T_h^{(3)} \leq t\right) = G_0(t) * G_1(t) * \cdots * G_n(t). \end{cases} \quad (2.2.36)$$

By combining equations (2.2.32), (2.2.35) and (2.2.36), we obtain the final expression :

$$E(Z_n^{(3)}(t)|Y_n^{(3)}) = Y_n^{(3)} \cdot [G_0(t) * G_1(t) * \cdots * G_{n-1}(t) - G_0(t) * G_1(t) * \cdots * G_n(t)] \quad (2.2.37)$$

where the symbol $*$ stands for the convolution. A better understanding of the appearance of the convolution lies in the following theorem:

Theorem 2.2.4. *If $X_0, X_1, \dots, X_n$ are independent random variable with the distribution functions $F_1, F_2, \dots, F_n$, then probability distribution function of the random variable $X_0 + X_1 + \cdots + X_n$ is given by $F_1 * F_2 * \cdots * F_n$.*

Since we calculate the sum of the lifetimes of a cell and its ancestors, by considering the lifetimes of cells as independent random variable and by applying (Theorem 2.2.4), the appearance of convolution is justified.

Assuming that, apart from the first generation, all other generations have the same distribution of times to division $G(t)$, equation (2.2.35) becomes

$$E(Z_n^{(3)}(t)|Y_n^{(3)}) = Y_n^{(3)} \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n}(t)]. \quad (2.2.38)$$

Let us now compute the mean of cells both alive or dead in the n th generation $E(Y_n^{(3)})$. Since we are dealing with discrete random variables, this refers to the embedded Galton-Watson process. Therefore, considering any Galton-Watson process described by the random vector $X = (X_0, X_1, \dots, X_n)$, if $X_0 = 1$ and $E(X_1) = \mu$, then $E(X_n) = \mu^n$ (see PP. 4, Chap. 1 [21]). Therefore, in order to know the value of $E(Y_n^{(3)})$, we merely need to compute $E(Y_1^{(3)})$ since we have the required the conditions.

Given that $E(Y_1^{(3)}) = 2P_3$, we have

$$E(Y_n^{(3)}) = (2P_3)^n, \quad (2.2.39)$$

and since we start with $N^{(3)}(0)$ cells of type 3, the expectations number of cells of this type is

$$\begin{aligned} N_n^{(3)}(t) &= N^{(3)}(0) \cdot E(Z_n^{(3)}(t)) \\ &= N^{(3)}(0) \cdot E[E(Z_n^{(3)}(t)|Y_n^{(3)})] \\ &= N^{(3)}(0) \cdot E[Y_n^{(3)} \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n}(t)]] \\ &= N^{(3)}(0) \cdot E(Y_n^{(3)}) \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n}(t)] \\ &= N^{(3)}(0) \cdot (2P_3)^n \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n}(t)]. \end{aligned} \quad (2.2.40)$$

The same idea may also be used for cell of types 1 and types 2 under assumptions related to each type. For instance, type 3 cells are ancestors of type 1 and 2 cells. Therefore, we obtain type 1 cells the general formula for the expected number of cells in the n th generation at time t :

$$N_n^{(1)}(t) = N^{(3)}(0) \cdot (2P_3)^n \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n-1}(t) * D(t)]. \quad (2.2.41)$$

For a better understanding of this expression, let us recall the fact that type 1 cell in the n th generation still and remains alive in the n th generation at time t only when the sum of lifetimes of its ancestors is less or equal to t and the sum of lifetimes of its ancestors and its own is greater than t .

Now, regarding type 2 cells, the assumptions are a little different. Indeed, a cell of type 2 in the n th generation still and remains alive in the n th generation at time t only when the sum of its ancestors lifetimes is less or equal to t . Hence, the expected numbers of cells of type 2 in the n th generation at time t is given by:

$$N_n^{(2)}(t) = N^{(3)}(0) \cdot (2P_3)^n \cdot [G_0(t) * G^{*(n-1)}(t)]. \quad (2.2.42)$$

Denote by $N_n^{(-1)}(t)$ the expectation number of died cells in the n th generation at time t . The cell is died at time t in the n th generation if the length of lifetimes of its ancestors and its own is less or equal to t . Remark for $n = 0$, died cells should come from type 1 and for $n \geq 1$ all the ancestors are cell of type 3. Thus, we have

$$N_0^{(-1)}(t) = N^{(1)}(0) \cdot D_0(t), \quad (2.2.43)$$

and for $n \geq 1$,

$$N_n^{(-1)}(t) = N^{(3)}(0) \cdot (2P_1)^n \cdot [G_0(t) * G^{*(n-1)}(t)]. \quad (2.2.44)$$

Before write the recursive relation which governs the expectation number of cells of each type, let us recall that the convolution of order 0 of a given distribution $F(t)$ is equal to 1. Therefore, by combining the equations (2.2.40)-(2.2.44), we obtain

$$\left\{ \begin{array}{l} N_0^{(1)}(t) = N^{(1)}(0) \cdot (1 - D_0(t)) \\ N_0^{(2)}(t) = N^{(2)}(0) \\ N_0^{(3)}(t) = N^{(3)}(0) \cdot (1 - G_0(t)) \\ N_0^{(-1)}(t) = N^{(1)}(0) \cdot D_0(t) \\ N_n^{(1)}(t) = N^{(3)}(0) \cdot (2P_1)^n \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n-1}(t) * D(t)] \\ N_n^{(2)}(t) = N^{(3)}(0) \cdot (2P_2)^n \cdot [G_0(t) * G^{*(n-1)}(t)] \\ N_n^{(3)}(t) = N^{(3)}(0) \cdot (2P_3)^n \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n}(t)] \\ N_n^{(-1)}(t) = N^{(3)}(0) \cdot (2P_1)^n \cdot [G_0(t) * G^{*(n-1)}(t)]. \end{array} \right. \quad (2.2.45)$$

This classification of cells into three subtypes present an ambiguity, due to the fact that in reality the term *quiescent cell* does have any specific connotation. Indeed, a cell's fate is one of the two following events: either the cell dies or it proliferates. For instance, one may ask the type of a cell which is born before the end of the experiment? The answer to this question can not be precise nor concise. The cell can be classified in the family of cells of type 2 which is not necessary true and in this case does not fit the basic assumptions of the model. Therefore, Miao et al. [6] suggests to simply consider two subtypes for the case when the number of quiescent cells is not large: cells that will divide (type 3) and cells that will die (type 1); therefore, we have $P_1 + P_3 = 1$. Hence, (2.2.45) becomes

$$\left\{ \begin{array}{l} N_0^{(1)}(t) = N^{(1)}(0) \cdot (1 - D_0(t)) \\ N_0^{(3)}(t) = N^{(3)}(0) \cdot (1 - G_0(t)) \\ N_0^{(-1)}(t) = N^{(1)}(0) \cdot D_0(t) \\ N_n^{(1)} = N^{(3)}(0) \cdot [2(1 - P_3)]^n \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n-1}(t) * D(t)] \\ N_n^{(3)}(t) = N^{(3)}(0) \cdot (2P_3)^n \cdot [G_0(t) * G^{*(n-1)}(t) - G_0(t) * G_1(t) * G^{*n}(t)] \\ N_n^{(-1)}(t) = N^{(3)}(0) \cdot [2(1 - P_3)]^n \cdot [G_0(t) * G^{*(n-1)}(t)]. \end{array} \right. \quad (2.2.46)$$

The system of equations that we have obtained and presented in (2.2.46) is less accurate than the one in (2.2.45) since it is a simplification of (2.2.45). When the number of quiescent cells is large, (2.2.46) fails to determine the distribution as (2.2.46). Since the lifetime of cells are continuous, any continuous distribution such as normal distribution, lognormal distribution, gamma distribution, exponential distribution, etc. can be used to approximate the process. Now the question is what is the adequate estimator for the distribution of time to death $D_i(t)$ and the time to division $G_i(t)$. The answer to this question has been discussed by Hyrien et al. [18] and be consider by Miao et al. [6], where the gamma distribution is chosen to be the best approximation in both cases. This choice is justified by the flexibility of the shape of gamma distribution.

After reviewing the existing models in [6], ones can ask about how much these models reflect the reality. To answer this question, we develop in the next session an algorithm simulation to test the accuracy of two particular models which are Smith-Martin Model and Cyton Model.

3. Results

This chapter is devoted to a detailed presentation of our main achievements. We describe the algorithm used to simulate a CFSE-like experiment and generate data. To facilitate the numerical implementation of the Cyton Model, an analytic formula for the rates of cell division and death is established. In the same vein, the number of cells for each generation at a given time t is written down through an analytic formula incorporating the incomplete gamma function. Graphs comparing simulated CSFE data and data provided by the models are depicted. A powerful tool from statistical analysis, namely the Average Relative Error (ARE), is used to pinpoint the models' accuracy.

3.1 Simulation

In the previous chapter, we reviewed four specific mathematical models for T cells proliferation as anticipated by Miao et al. [6]. Namely, the Smith-Martin model, the Linear Birth-death ODE Model, the Cyton Model, and the Age-Dependent Branching Process. The first model, viewed as a deterministic model (no random event involved), is presented in a DDE form and differs from the others that are stochastic (incorporating the probabilistic structure for cell division and death) and combines the compartmental structure analysis. It would be interesting to compare the models in order to determine/detect their weaknesses and strengths. This could be a vast, tedious and incommensurate time-consuming task. In the frame of this work, we focus our attention on only two specific models: the Smith-Martin and Cyton models by comparing and testing their accuracy. From a biological perspective, the Cyton Model is the most interesting of all the models that we reviewed. What is really important from a biological perspective is not the fancifulness of the mathematics used to construct a model but the depth of the biological insight that the model provides. For this reason, the Cyton Model is much more interesting than the more complicated Age-Dependent Multitype Branching Process Model. In fact, the Cyton Model was recently used by Marchingo et al. [34] to obtain what we can consider to be a landmark result in the field of immunology: it was used to discover a simple calculus that T cells perform when they respond to different types of signals.

We simulate data as a CFSE labeling experiment could do, by using the *individual based-simulation* (IBS) technique. The IBSs known also as agent based simulations, is based on global consequences of local interactions of a population's members. These processes typically consist of a framework in which interactions occur between some number of individuals defined in terms of their behaviors (procedural rules) and characteristic parameters. In an individual-based model, the characteristics of each individual are tracked through time. This method, though stochastic, contrast with other modeling techniques where the characteristics of the population are averaged together and the model attempts to simulate changes in these averaged characteristics for the whole population [35]. Several authors have used IBS techniques to simulate cell dynamics Miao et al. [6], Byrne and Drasdo [36], Walker and Southgate [37], Bogle and Dunbar [38]. The interest of using IBS in this work lies in the fact that T cell's proliferation is a complex system (each cell has its own unique experience, the probabilistic effects enter at every stage, the fate of cells are no longer

correlated...). In order to achieve our goal, the simulation protocol used throughout the chapter is divided into three steps as listed below:

Step 1: initialization step, which contains the following sub-steps:

1. initialization of parameters for cell dynamics: proliferation rate and death rates for different division classes of cells,
2. initialization of a container to pour the number of cells in different division classes at different time points,
3. initially, at time point 1, we have N_0 cells in the 0th division class, and zero cells in all the other classes. We randomly select a proportion $P_a \cdot (1 - P_{ad})$ of these cells that will be destined to divide. We continue by keeping track the fate of $N_0 \cdot P_a \cdot (1 - P_{ad})$ cells only. Here, P_a is the probability of a cell to be activated, whereas P_{ad} represents the probability of cell death after being activated.
4. We define a vector containing the duration of the different time points.
5. We determine the duration of our simulation.

Step 2:

1. We identify all events that can occur in the current time point. For example, in time point 1, there are on average $N_0 \cdot P_a \cdot (1 - P_{ad})$ cells, each of which can either divide or die. Therefore, there are on average $2 \cdot N_0 \cdot P_a \cdot (1 - P_{ad})$ possible events that can occur in time point 1.
2. We randomly sample the waiting time for each event to occur from the appropriate distributions, which in our case, is the exponential distributions.
3. We identify the event with the shortest sampled waiting time and allow it to occur. We then update the age structure of the cell population accordingly. For instance, if a cell that is currently in the 0th division class has the shortest calculated waiting time to divide, then the number of cells in that class should decrease by 1 while the number of cells in the next division class should increase by either 2 (with probability P_3^2) or 1 (with probability $2P_3$). On the other hand, if a cell in the 0th division class dies, then the number of cells in that class will simply decrease by 1.
4. We add the waiting time of the event that just occurred (i.e. the smallest of the calculated waiting times) to the time points vector.

Step 3:

If the sum of the entries in the time points vector is greater than or equal to the final time, then stop the simulation. Otherwise, return to step 2.

To properly achieve this simulation, we used the following parameter values: $N_0 = 50000$, $p_a = 1$, $p_{ad} = 0.50$, $p_1 = 0.50$, $p_3 = 0.5$. For the distribution time of proliferation and death, we use

the exponential distributions with parameter $\mu_{p0} = 40$ for waiting time to proliferate of the 0th generation, and $\mu_p = 8$ for others generations. Concerning the time to death, we use $\mu_{d0} = 60$ for the 0th generation and $\mu_d = 60$ for others. These steps allow us to have outputs that are data contain the number of cells of different generations that are going to proliferate or die at different time points and that are respectively denoted as $N_iP(t)$ and $N_id(t)$ with $i = 0, 1, \dots$. Note that the total number of cells of i th generation at time t is $N_i(t) = N_iP(t) + N_id(t)$. For this simulation, we produce three data sets that are purposely used to meet our target.

After designing the simulation, we mainly fit two models to the simulated outputs.

3.2 Further simplifications of the Cyton Model

In the process of using the Cyton Model to fit CFSE data, it was discovered early on that this process takes a lot of time up to several hours. We have since found it adequate to simplify the Cyton Model such that the running time could be significantly reduced. To this end, we have restricted our investigations to the case where the distributions of time to divide and the time to die follow exponential law. Furthering the simplifications, a formula for the rate of cells that are going to proliferate and die, regardless of the values of progressor fraction, is obtained and described in detail.

As a starting point, let us rewrite the Cyton Model for the rates $r_n^{div}(t)$ and $r_n^{die}(t)$ of cell's division and death respectively and at n th generation at time t for $f_n = 1$ ($n = 0, 1, 2, \dots$):

$$r_0^{div}(t) = N(0) \cdot \left(1 - \int_0^t \psi_0(\tau) d\tau\right) \cdot \phi_0(t), \quad (3.2.1)$$

$$r_0^{die}(t) = N(0) \cdot \left(1 - \int_0^t \phi_0(\tau) d\tau\right) \cdot \psi_0(t), \quad (3.2.2)$$

$$r_n^{div}(t) = 2 \cdot \int_0^t r_{n-1}^{div}(\tau) \cdot \left(1 - \int_0^{t-\tau} \psi_n(\zeta) d\zeta\right) \cdot \phi_n(t-\tau) d\tau, \quad (3.2.3)$$

and

$$r_n^{die}(t) = 2 \cdot \int_0^t r_{n-1}^{die}(\tau) \cdot \left(1 - \int_0^{t-\tau} \phi_n(\zeta) d\zeta\right) \cdot \psi_n(t-\tau) d\tau. \quad (3.2.4)$$

We further assume that the time to divide and the time to die follow exponential distributions defined as

$$\phi_n(t) = \begin{cases} \frac{1}{p_0} e^{-\frac{t}{p_0}}, & \text{for } n = 0 \\ \frac{1}{p} e^{-\frac{t}{p}}, & \text{for } n \geq 1 \end{cases} \quad \text{and} \quad \psi_n(t) = \begin{cases} \frac{1}{d_0} e^{-\frac{t}{d_0}}, & \text{for } n = 0 \\ \frac{1}{d} e^{-\frac{t}{d}}, & \text{for } n \geq 1. \end{cases} \quad (3.2.5)$$

Since $r_n^{die}(t)$ depends on $r_{n-1}^{div}(t)$, we compute some terms of $r_n^{div}(t)$ where $n \geq 0$, and later proceed by conjecturing the general expression which is finally proven by recurrence.

Let us note that,

$$1 - \int_0^t \phi_n(\tau) d\tau = \begin{cases} e^{-\frac{t}{p_0}}, & \text{for } n = 0 \\ e^{-\frac{t}{p}}, & \text{for } n \geq 1 \end{cases} \quad \text{and} \quad 1 - \int_0^t \psi_n(\tau) d\tau = \begin{cases} e^{-\frac{t}{d_0}}, & \text{for } n = 0 \\ e^{-\frac{t}{d}}, & \text{for } n \geq 1. \end{cases} \quad (3.2.6)$$

Under this consideration, for $n = 0$ we have

$$r_0^{div}(t) = N(0) \cdot \left(1 - \int_0^t \psi_0(\tau) d\tau\right) \cdot \phi_0(t) = \frac{N(0)}{p_0} \cdot e^{-\gamma_0 t}, \quad (3.2.7)$$

where γ_0 is given by

$$\gamma_0 = \frac{p_0 + d_0}{p_0 d_0}. \quad (3.2.8)$$

Likewise, we obtain

$$r_0^{die}(t) = N(0) \cdot \left(1 - \int_0^t \phi_0(\tau) d\tau\right) \cdot \psi_0(t) = \frac{N(0)}{d_0} \cdot e^{-\gamma_0 t}. \quad (3.2.9)$$

For $n = 1$, we have

$$\begin{aligned} r_1^{div}(t) &= 2 \cdot \int_0^t r_0^{div}(\tau) \cdot \left(1 - \int_0^{t-\tau} \psi_1(\zeta) d\zeta\right) \cdot \phi_1(t-\tau) d\tau \\ &= \frac{2N(0)}{p_0 p} \cdot e^{-\gamma t} \left[\frac{e^{-(\gamma_0 - \gamma)t} - 1}{\gamma - \gamma_0} \right], \end{aligned} \quad (3.2.10)$$

where

$$\gamma = \frac{p + d}{pd}. \quad (3.2.11)$$

For $n = 2$, we have

$$\begin{aligned} r_2^{div}(t) &= 2 \cdot \int_0^t r_1^{div}(\tau) \cdot \left(1 - \int_0^{t-\tau} \psi_2(\zeta) d\zeta\right) \cdot \phi_2(t-\tau) d\tau \\ &= \frac{2^2 N(0)}{p_0 p^2} \cdot e^{-\gamma t} \left[\frac{e^{-(\gamma_0 - \gamma)t}}{(\gamma - \gamma_0)^2} - \frac{1}{(\gamma - \gamma_0)^2} - \frac{t}{\gamma - \gamma_0} \right]. \end{aligned} \quad (3.2.12)$$

For $n = 3$, we have

$$\begin{aligned} r_3^{div}(t) &= 2 \cdot \int_0^t r_2^{div}(\tau) \cdot \left(1 - \int_0^{t-\tau} \psi_3(\zeta) d\zeta\right) \cdot \phi_3(t-\tau) d\tau \\ &= \frac{2^3 N(0)}{p_0 p^3} \cdot e^{-\gamma t} \left[\frac{e^{-(\gamma_0 - \gamma)t}}{(\gamma - \gamma_0)^3} - \frac{1}{(\gamma - \gamma_0)^3} - \frac{t}{(\gamma - \gamma_0)^2} - \frac{t^2}{2!(\gamma - \gamma_0)} \right]. \end{aligned} \quad (3.2.13)$$

Hence, for $n \geq 0$, our conjecture suggests that

$$\begin{cases} r_0^{div}(t) = \frac{N(0)}{p_0} \cdot e^{-\gamma_0 t} \\ r_n^{div}(t) = \frac{2^n N(0)}{p_0 p^n} \cdot e^{-\gamma t} \left[\frac{e^{-(\gamma_0 - \gamma)t}}{(\gamma - \gamma_0)^n} - \sum_{i=0}^{n-1} \frac{t^i}{i! (\gamma - \gamma_0)^{n-i}} \right], \quad n \geq 1. \end{cases} \quad (3.2.14)$$

Now let us confirm our conjecture by induction.

For $n = 1$, the formula trivially works. Let us suppose that our formula works up to an arbitrary $k \geq 1$, and show that it works for $k + 1$. In this scenario and in full detail, we have

$$\begin{aligned} r_{k+1}^{div}(t) &= 2 \cdot \int_0^t r_k^{div}(\tau) \cdot \left(1 - \int_0^{t-\tau} \psi_{k+1}(\zeta) d\zeta \right) \cdot \phi_{K+1}(t - \tau) d\tau \\ &= 2 \cdot \int_0^t \frac{2^k N(0)}{p_0 p^k} \cdot e^{-\gamma \tau} \left[\frac{e^{-(\gamma_0 - \gamma)\tau}}{(\gamma - \gamma_0)^k} - \sum_{i=0}^{k-1} \frac{\tau^i}{i! (\gamma - \gamma_0)^{k-i}} \right] e^{-\frac{t-\tau}{d}} \cdot \frac{e^{-\frac{t-\tau}{p}}}{p} d\tau \\ &= \frac{2^{k+1} N(0)}{p_0 p^{k+1}} \cdot e^{-\gamma t} \int_0^t e^{-\gamma \tau} \left[\frac{e^{-(\gamma_0 - \gamma)\tau}}{(\gamma - \gamma_0)^k} - \sum_{i=0}^{k-1} \frac{\tau^i}{i! (\gamma_0 - \gamma)^{k-i}} \right] e^{\gamma \tau} d\tau \\ &= \frac{2^{k+1} N(0)}{p_0 p^{k+1}} \cdot e^{-\gamma t} \left[\frac{e^{-(\gamma_0 - \gamma)t}}{(\gamma - \gamma_0)^{k+1}} - \sum_{i=0}^{k-1} \frac{\tau^{i+1}}{(i+1)! (\gamma - \gamma_0)^{k-i}} \right]_{\tau=0}^{\tau=t} \\ &= \frac{2^{k+1} N(0)}{p_0 p^{k+1}} \cdot e^{-\gamma t} \left[\frac{e^{-(\gamma_0 - \gamma)t}}{(\gamma - \gamma_0)^{k+1}} - \sum_{i=0}^k \frac{t^i}{i! (\gamma - \gamma_0)^{k-i+1}} \right], \end{aligned} \quad (3.2.15)$$

which at the end of investigations reveals that our initial hypothesis holds. This therefore confirms the validity of our formula (3.2.14). These results are an excellent starting point for reducing the running time for the Cyton model as pointed out below.

Now that we possess the general formula for $r_n^{div}(t)$, from equations (3.2.4), (3.2.6) and (3.2.9), one can compute a formula for $r_n^{die}(t)$, $n \geq 0$. Thus,

$$\begin{cases} r_0^{die}(t) = \frac{N(0)}{d_0} \cdot e^{-\gamma_0 t} \\ r_n^{die}(t) = \frac{2^n N(0)}{p_0 p^{n-1} d} \cdot e^{-\gamma t} \left[\frac{e^{-(\gamma_0 - \gamma)t}}{(\gamma - \gamma_0)^n} - \sum_{i=0}^{n-1} \frac{t^i}{i! (\gamma - \gamma_0)^{n-i}} \right] = \frac{p}{d} \cdot r_n^{div}(t), \quad n \geq 1. \end{cases} \quad (3.2.16)$$

At this stage, can compute the number of cells in each generation at time t . From equation (2.2.23), we can deduce that

$$N_0(t) = N(0) e^{-\gamma_0 t}, \quad (3.2.17)$$

and for $n \geq 1$

$$\begin{aligned}
N_n(t) &= \int_0^t \frac{2^n}{p_0 p^{n-1}} \cdot N(0) e^{-\gamma\tau} \left\{ \left(\frac{e^{-(\gamma_0-\gamma)\tau}}{(\gamma-\gamma_0)^{n-1}} - \sum_{i=0}^{n-2} \frac{t^i}{i!(\gamma-\gamma_0)^{n-i-1}} \right) - \right. \\
&\quad \left. \gamma \cdot \left(\frac{e^{-(\gamma_0-\gamma)\tau}}{(\gamma-\gamma_0)^n} - \sum_{i=0}^{n-1} \frac{t^i}{i!(\gamma-\gamma_0)^{n-i}} \right) \right\} d\tau \\
&= \frac{2^n}{p_0 p^{n-1}} \cdot N(0) \left[\int_0^t \frac{e^{-\gamma_0\tau}}{(\gamma-\gamma_0)^{n-1}} d\tau - \sum_{i=0}^{n-2} \frac{t}{i!(\gamma-\gamma_0)^{n-i-1}} \int_0^t \tau^i e^{-\gamma\tau} d\tau \right. \\
&\quad \left. - \gamma \left(\int_0^t \frac{e^{-\gamma_0\tau}}{(\gamma-\gamma_0)^n} d\tau - \sum_{i=0}^{n-1} \frac{t}{i!(\gamma-\gamma_0)^{n-i}} \int_0^t \tau^i e^{-\gamma\tau} d\tau \right) \right] \\
&= \frac{2^n}{p_0 p^{n-1}} \cdot N(0) \left[\frac{1}{\gamma_0(\gamma-\gamma_0)^{n-1}} (1 - e^{-\gamma_0 t}) - \sum_{i=0}^{n-2} \frac{\Gamma(i+1, \gamma t)}{i!(\gamma-\gamma_0)^{n-i-1} \gamma^{i+1}} \right. \\
&\quad \left. - \gamma \left(\frac{1}{\gamma_0(\gamma-\gamma_0)^n} (1 - e^{-\gamma_0 t}) - \sum_{i=0}^{n-1} \frac{\Gamma(i+1, \gamma t)}{i!(\gamma-\gamma_0)^{n-i} \gamma^{i+1}} \right) \right], \tag{3.2.18}
\end{aligned}$$

where $\Gamma(s, x)$ is the lower incomplete gamma function defined by

$$\Gamma(s, x) = \int_0^x \xi^{s-1} e^{-\xi} d\xi. \tag{3.2.19}$$

It is worth noting that for the case when the progressor fraction is not necessary equal to 1 a general formula can always be performed for r_n^{div} . Indeed, by realizing that after each integration step, this factor always stands out of the integral sign, r_n^{div} , $n \geq 0$ is obtained from (3.2.14) by multiplying the first equation by f_0 and the second one by $f_1 f_2 \cdots f_n$. The result of these investigations reads:

$$\begin{cases} r_0^{div}(t) = \frac{f_0 N(0)}{p_0} \cdot e^{-\gamma_0 t} \\ r_n^{div}(t) = \frac{2^n N(0)}{p_0 p^n} \cdot \left(\prod_{i=0}^n f_i \right) \cdot e^{-\gamma t} \left[\frac{e^{-(\gamma_0-\gamma)t}}{(\gamma-\gamma_0)^n} - \sum_{i=0}^{n-1} \frac{t^i}{i!(\gamma-\gamma_0)^{n-i}} \right], \quad n \geq 1. \end{cases} \tag{3.2.20}$$

Then rate of cells that are going to die, regardless the progressor fraction, can be obtained from (2.2.22) and (3.2.20).

First let us remark that

$$1 - f_n \int_0^{t-\tau} \phi(\zeta) d\zeta = 1 - f_n + f_n e^{-\frac{t-\tau}{p}}, \tag{3.2.21}$$

then,

$$\begin{aligned}
r_n^{die}(t) &= 2 \cdot \int_0^t \frac{2^{n-1} N(0)}{p_0 p^{n-1}} \cdot \left(\prod_{i=0}^{n-1} f_i \right) \cdot e^{-\gamma \tau} \left[\frac{e^{-(\gamma_0 - \gamma)\tau}}{(\gamma - \gamma_0)^{n-1}} - \sum_{i=0}^{n-2} \frac{\tau^i}{i! (\gamma - \gamma_0)^{n-i-1}} \right] \\
&\quad \left(1 - f_n \int_0^{t-\tau} \frac{1}{p} e^{\zeta/p} d\zeta \right) \frac{1}{d} e^{-\frac{t-\tau}{d}} d\tau \\
&= \frac{2^n N(0)}{p_0 p^{n-1} d} \cdot \left(\prod_{i=0}^{n-1} f_i \right) \cdot \int_0^t \left(\frac{e^{-\gamma_0 \tau}}{(\gamma - \gamma_0)^{n-1}} - \sum_{i=0}^{n-2} \frac{\tau^i e^{-\gamma \tau}}{i! (\gamma - \gamma_0)^{n-i-1}} \right) \left[(1 - f_n) + \right. \\
&\quad \left. f_n e^{-\frac{t-\tau}{p}} \right] e^{-\frac{t-\tau}{d}} d\tau \\
&= \frac{2^n N(0)}{p_0 p^{n-1} d} \cdot \left(\prod_{i=0}^{n-1} f_i \right) \cdot \left\{ (1 - f_n) e^{-t/d} \left[\int_0^t \frac{e^{-(\gamma_0 - \frac{1}{d})\tau}}{(\gamma - \gamma_0)^{n-1}} - \sum_{i=0}^{n-2} \frac{1}{i! (\gamma - \gamma_0)^{n-1-i}} \right. \right. \\
&\quad \left. \left. \int_0^t \tau^i e^{-(\gamma - \frac{1}{d})\tau} d\tau \right] + f_n e^{-\gamma t} \left[\int_0^t \frac{e^{-(\gamma_0 - \gamma)\tau}}{(\gamma - \gamma_0)^{n-1}} - \sum_{i=0}^{n-2} \frac{1}{i! (\gamma - \gamma_0)^{n-1-i}} \int_0^t \tau^i d\tau \right] \right\} \\
&= \frac{2^n N(0)}{p_0 p^{n-1} d} \cdot \left(\prod_{i=0}^{n-1} f_i \right) \cdot \left\{ (1 - f_n) e^{-t/d} \left[\frac{e^{-(\gamma_0 - \frac{1}{d})t}}{(\frac{1}{d} - \gamma_0)(\gamma - \gamma_0)^{n-1}} - \right. \right. \\
&\quad \left. \left. \sum_{i=0}^{n-2} \frac{\Gamma[i+1, (\gamma - \frac{1}{d})t]}{i! (\gamma - \gamma_0)^{n-1-i} (\gamma - \frac{1}{d})^{i+1}} \right] + f_n e^{-\gamma t} \left[\frac{e^{-(\gamma_0 - \gamma)t} - 1}{(\gamma - \gamma_0)^n} - \sum_{i=0}^{n-2} \frac{t^{i+1}}{(i+1)! (\gamma - \gamma_0)^{n-1-i}} \right] \right\} \\
&= \frac{2^n N(0)}{p_0 p^{n-1} d} \cdot \left(\prod_{i=0}^{n-1} f_i \right) \cdot \left\{ (1 - f_n) \left[\frac{e^{-\gamma_0 t}}{(\frac{1}{d} - \gamma_0)(\gamma - \gamma_0)^{n-1}} - \sum_{i=0}^{n-2} \frac{e^{-\frac{t}{d}} \Gamma[i+1, (\gamma - \frac{1}{d})t]}{i! (\gamma - \gamma_0)^{n-1-i} (\gamma - \frac{1}{d})^{i+1}} \right] \right. \\
&\quad \left. + f_n \left[\frac{e^{-\gamma_0 t} - e^{-\gamma t}}{(\gamma - \gamma_0)^n} - \sum_{i=0}^{n-2} \frac{e^{-\gamma t} t^{i+1}}{(i+1)! (\gamma - \gamma_0)^{n-1-i}} \right] \right\}. \tag{3.2.22}
\end{aligned}$$

Therefore, the rate of cell that are going to divide for the n th generation, $n = 0, 1, 2 \dots$ is given by:

$$\begin{cases} r_0^{die} = \frac{N(0)}{d} \left[(1 - f_0) e^{-t_0/d} + f_0 e^{-\gamma_0 t} \right] \\ r_n^{die} = \frac{2^n N(0)}{p_0 p^{n-1} d} \cdot \left(\prod_{i=0}^{n-1} f_i \right) \cdot \left\{ (1 - f_n) \left[\frac{e^{-\gamma_0 t}}{(\frac{1}{d} - \gamma_0)(\gamma - \gamma_0)^{n-1}} - \sum_{i=0}^{n-2} \frac{e^{-\frac{t}{d}} \Gamma[i+1, (\gamma - \frac{1}{d})t]}{i! (\gamma - \gamma_0)^{n-1-i} (\gamma - \frac{1}{d})^{i+1}} \right] \right. \\ \left. + f_n \left[\frac{e^{-\gamma_0 t} - e^{-\gamma t}}{(\gamma - \gamma_0)^n} - \sum_{i=0}^{n-2} \frac{e^{-\gamma t} t^{i+1}}{(i+1)! (\gamma - \gamma_0)^{n-1-i}} \right] \right\}. \end{cases} \tag{3.2.23}$$

3.3 Analysis of Smith-Martin Model and Cyton Model

In order to fit our mathematical models to CFSE data from individual-based simulations, we start the simulation with 50000 cells of age zero (i.e., cells that have undergone zero division). We define the monitoring times (time points) as 12 hours (i.e. after each 12 hours, we evaluate the number of cells present in each generation). For each model, we define a range where the parameters will be picked. Parameters for Smith-Martin Model are the rate of division for the 0th generation λ_0 , the rate of division for others generations λ , the death rate for the 0th generation d_0 , the death rate for other generations d and the delay time Δ . For Cyton Model, parameters are: progressor fractions f_0 and f for 0th and other generations respectively, the waiting times to divide μ_{p_0} and μ_p , and the waiting times to die μ_{d_0} and μ_d .

For each real data set from individual data based simulation, we run one hundred simulations from the different models, and select the optimal parameters (i.e the parameters which minimize the sum of square of differences (SSD)). Among those sets of parameters, only one set is considered (the set with the lower SSD) be replaced in the model to generate the data which will be plotted against the average of three data from individual-based simulations. This process is applied to the two models that we check for accuracy. A question about the optimality of the estimated parameters may arise: do the parameters represent a local minimum of SSD, since the minimum depends on the starting point since we are using the deterministic search algorithm? The answer to this question resides in the number of simulations that are run. Given that we run one hundred simulations before choosing the optimal parameters and the fact that the starting point is chosen in a random fashion, we may assert that somehow the SSD is a global minimum. After fitting the models to simulated data, we measure their accuracy (see figures 3.1 and 3.2).

Figures 3.1 and 3.2, reveal that the Cyton Model better fits real data from the 0th to the 8th generation as real and numerical data are in adequately good agreement. Indeed, it can be easily observed that the cells' difference in absolute value (i.e. the shift between real and numerical data at the same time point) at the same time points in the same generation is on average smaller in the Cyton Model as compared with the Smith-Martin Model. This is basically a naïve conclusion arising from a global observation of curves. A local comparison of graphs taken individually can also be done. In what follows, a remarkable comparative analysis is performed by introducing a powerful statistical tool: the Average Relative Error (ARE). This tool gives us the possibility of comparing the parameters that the two models of interest here share.

It is relevant for further purposes to note that the two models only share four comparable parameters, the first pair of which maps the division and the second pair maps the death. The rates of cells which leave the A phase to B phase in the Smith-Martin Model (namely λ_0 and λ) are respectively comparable with the waiting times to divide in the Cyton Model (μ_{p_0} and μ_p namely). It should also be noted that μ_{p_0} and μ_p represent the parameters of distribution of time to divide (in our case, an exponential distribution). In the same vein, the death rates d_0 and d in the Smith-Martin Model are comparable with the waiting time μ_{d_0} and μ_d in the Cyton Model. Therefore, one of the best ways to check the accuracy of our models consists of measuring how close they are two models in experiments. For this purpose, we will compute ARE between the comparable parameters of each model and the corresponding comparable values that are used for individual-based simulation.

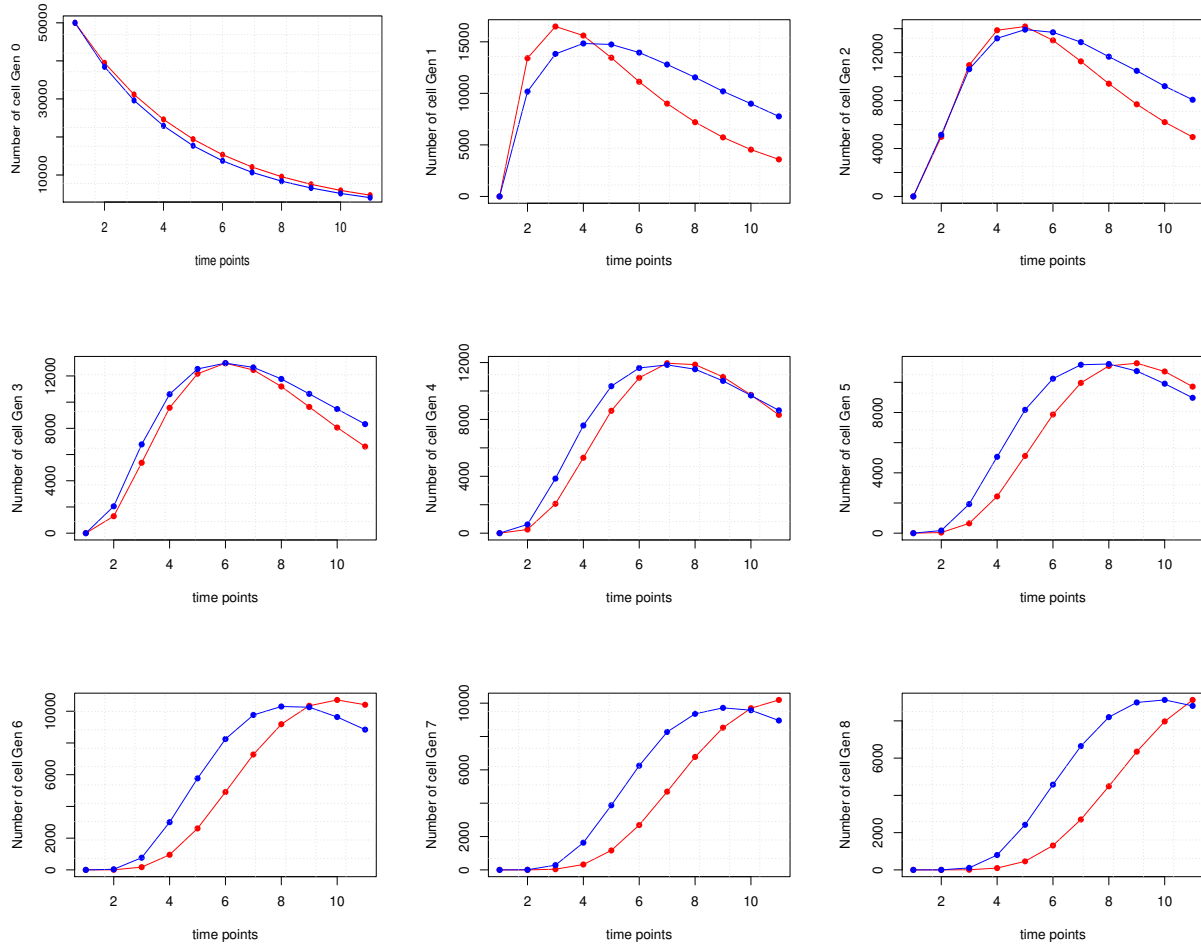


Figure 3.1: Fitting of the **Smith-Martin Model** with **real data**. Number of cells versus time points from the 0th to the 8th generation. Blue lines are real data generated from an individual-based simulation while red lines are data generated from the Smith-Martin model.

As already mentioned, ARE allows us to evaluate which of the models is the best. Note that the model with the parameters with smaller ARE is the best. ARE is calculated for the following parameters:

- the waiting times to divide for cells in 0th generation denoted as μ_{p_0} ,
- the waiting times to divide for cells in n th generation ($n \geq 1$) denoted as μ_p ,
- the waiting times to die for cells in 0th generation denoted as μ_{d_0} ,
- the waiting times to die for cells in n th generation ($n \geq 1$) denoted as μ_d .

In Smith-Martin Model, the waiting time for cells to proliferate is equal to the time to leave the

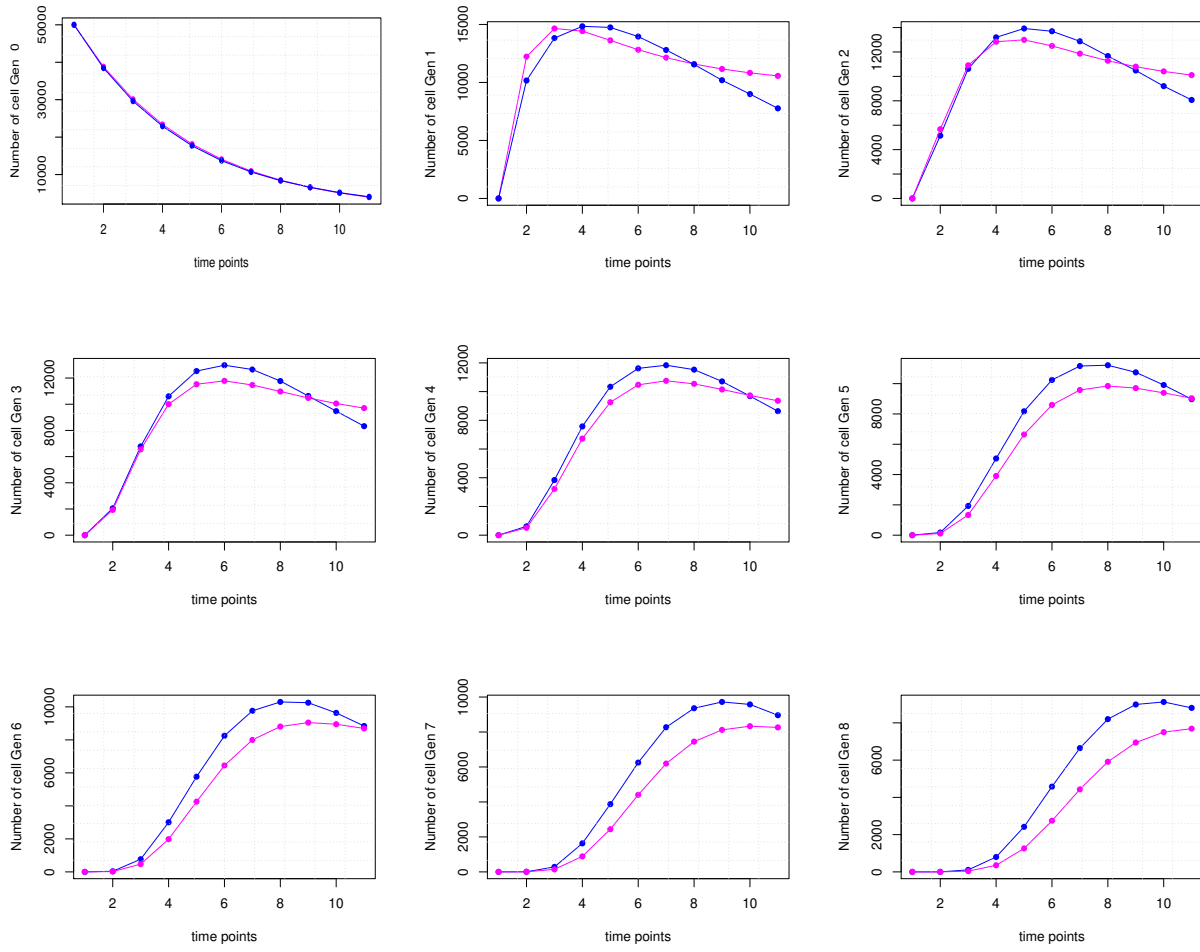


Figure 3.2: Fitting **Cyton Model** with **real data**. Number of cells versus time points from the 0th to the 8th generation. Blue lines are real data generated from an individual-based simulation while magenta lines are data generated from the Cyton model.

A phase to the B phase plus the length of the B phase. Therefore, for the 0th generation,

$$\mu_{p0} = \frac{1}{\lambda_0} + \Delta, \quad (3.3.1)$$

where $1/\lambda_0$ is the time to leave the A phase to B phase in generation 0. The waiting time to proliferate for others generations is given by

$$\mu_p = \frac{1}{\lambda} + \Delta, \quad (3.3.2)$$

where $1/\lambda$ stands for the time to leave the A phase to B phase for others generations $n \neq 0$. Regarding the waiting times to die in the same model, they are respectively given by

$$\mu_{d0} = \frac{1}{d_0}, \quad (3.3.3)$$

for generation 0 and

$$\mu_d = \frac{1}{d}, \quad (3.3.4)$$

for other generations.

The default parameters used to fit the models range in an order of $\pm 10\%$ between those used to generate the data set namely $\mu_{p0} = 40$ hours, $\mu_{d0} = 60$ hours, $\mu_p = 8$ hours and $\mu_d = 60$. This range of fluctuation enables us to account for perturbations. Previous works such as Miao et al. [39] and Miao et al. [6] have revealed that ARE is a powerful instrument to check the accuracy of models. A simple description of ARE is presented below:

Let θ_j be the j th parameter that we are willing to estimate; the formula for ARE (denoted here as ARE_j) is given by

$$ARE_j = \frac{1}{N} \sum_{i=0}^N \frac{|\theta_j - \theta_{ji}|}{|\theta_j|} \times 100\%, \quad (3.3.5)$$

where N represents the number of data sets from individual-based simulations. For instance, an analytic expression for ARE associated with a waiting time to divide for cells in the 0th generation for the Smith-Martin Model reads:

$$ARE_{\mu_{p0}} = \frac{1}{3} \left(\frac{|\mu_{p0} - \hat{\mu}_{p01}|}{|\mu_{p0}|} + \frac{|\mu_{p0} - \hat{\mu}_{p02}|}{|\mu_{p0}|} + \frac{|\mu_{p0} - \hat{\mu}_{p03}|}{|\mu_{p0}|} \right) \times 100\%, \quad (3.3.6)$$

where $\mu_{p0} = 1/\lambda_0 + \Delta$, while μ_{p0i} , ($i = 1, 2, 3$) is the best approximations from each data set i from individual-based simulation. By using the same idea for waiting time to divide of n th ($n \geq 1$), waiting time to die of 0th and waiting time to die of n th ($n \geq 1$) and using numerical values, we obtain the following table:

Table 3.1: Average Relative Error (ARE) in the models fitting cell counts and simulated data.

Model	Parameter range	ARE(%) of waiting time to division of gen.0	ARE(%) of waiting time to division of gen. ≥ 1	ARE(%) of waiting time to death of gen.0	ARE(%) of waiting time to division of gen. ≥ 1
Smith-Martin	$\pm 10\%$	22.54	67.51	94	128.53
Cyton	$\pm 10\%$	18.36	49.25	91.34	175.80

The ARE comparison shows us which model helps us to better approximate the parameters which govern T cells' proliferation. So, from ARE we can conclude which model is more accurate. From table 3.1, we can see that concerning the waiting time to divide and die for the 0th generation cells, the AREs of Cyton Model is smaller than that of the Smith-Martin Model (18.36% vs 22.54% for waiting time to divide and 91.34% vs 94% for waiting time to die) but the difference is not big. This means that for the cells that undergo 0 division, there is not huge distance between the two models for the approximation of waiting time to divide, but the Cyton is the best. Concerning cells of other generations, we can see that that the ARE waiting time to divide of the Cyton Model is still the best model to approximate (49.25% vs 67.52%). Smith-Martin Model tend to be worst comparing to the case of the 0th generation. Concerning the waiting time to die for cells which undergo at least one division, the Smith-Martin Model performs better than the Cyton Model (128.53% vs 175.80%). Therefore, we can conclude that the Cyton Model is better than the Smith-Martin Model for estimating the parameters of T cells' proliferation. The calculation of ARE for different estimated parameters has enabled us to conclude on the quality of the two models considered. A question may arise about the different values of ARE. Is it really the best value that we can obtain by using the mathematical model to estimate the parameters? Can we qualitatively improve the Smith-Martin Model to render it better to approximate those parameters? Attempts to answers these questions are presented in the next chapter.

4. Discussion and Conclusion

The CFSE labeling technique has been an important method to quantify the cells' dynamics. However, the experimental data produced by this technique does not provide direct information about the parameters that govern T-cell proliferation dynamics. To estimate the parameters governing this dynamic, we have initially revisited the experimental protocol for monitoring the dynamics of T lymphocytes using the CFSE dye as mentioned by Quah et al. [4] and Quah and Parish [5], which allowed us to follow the fate of a cell until the 8th generation. This understanding of the experimental process has also allowed us to better understand the context and assumptions adopted for constructing mathematical models. Thereafter, we revisited some mathematical models that have served to interpret the CFSE labeling experiment.

We began with the simplest model which divides the cell's cycle into two phases: one phase with a defined duration (B phase), and an undefined duration phase (A phase). This model also known as Smith-Martin Model is written as a set of coupled delayed differential equations describing the change of the cell population of each generation over time. By considering other assumptions such as the slow cell division, the A phase becomes negligible compared to the total duration of the cell cycle. Then, the Smith-Martin Model can reduce to a simpler ordinary differential equation model known as Linear Birth-death Model. Having noticed some limits of the deterministic approach, which is the lack of information on cell death, we consider the Cyton Model which combines a compartmental approach and probabilistic structure, where the time to division and death are defined by probability distributions. This model proves to be more realistic than the first. However, it presents some defects as mentioned by Hyrien et al. [18] that division probability is entirely determined by the distribution of time to division. Following this, we propose the study of a more stochastic suitable model known as Age-Dependent Multitype Branching Process, which is a robust model built on assumptions that make it close to a realistic situation as described by the CFSE labeling experiments.

After having revisited the mathematical models, we used an individual-based simulation reflecting the real experiments to investigate how well the Smith Martin and Cyton models estimate the parameters that govern T-cell proliferation dynamics. We produced three datasets using the individual-based simulation and then fitted the models to the simulated data. The analysis allowed us to show that Cyton Model is more accurate than the Smith-Martin Model. However, an analysis of the Age-Dependent Multitype Branching Process may provide us with more specific and precise information on these parameters. But, it is of major importance to note that a previous study by Miao et al. [6] has shown that the Age-Dependent Multitype Branching Process model is slightly better than Cyton and Smith-Martin models to approach the parameters that govern the dynamics of T cells' proliferation. This is done by assuming the distributions of time division and death to be approximated by a gamma distribution. However, the Age-Dependent Multitype Branching Process is a complicated model difficult to handle. This explains why studying the Cyton Model is captivating since it has been shown that it is comparable with the Age-Dependent Multitype Branching Process [18].

As seen earlier, and based on some studies performed by Hawkins et al. [14], and later by Wellard et al. [16] on the Cyton Model, and the events of division and death can be treated as

competing random events that follow continuous distributions. The fate of the cell is determined by which event occurs first. But in our study, we have considered that after the first division, all "division" events are exponentially distributed with the parameter $1/p$ and all "death" events are also exponentially distributed with the parameter $1/d$. This implies that after the 0th generation, all cells, regardless of the generation, have similarly behavior. This assumption seems to be less valid. But we use it here to reduce the cost of computations and the complexity of the model. For future studies, it will be interesting to consider the case when the probability distributions characterizing division and death depend on the generation. In the Smith-Martin Model we saw that there is a stochastic recruitment from the A phase to the B phase due to the fact that the length of the A phase is undetermined. The waiting time in the A phase is exponentially distributed with the parameter λ . But the Smith-Martin model does not allow us to better understand the process of cell death. For instance, Miao et al. [6] considered the death rate after the 0th generation to be the same, which rendered the model less informative on cell death. So, as mentioned earlier by Ganusov et al. [40], to ameliorate the Smith-Martin Model, more attention should be paid to cell death.

We end this essay by pointing out some advantages and disadvantages of the Smith-Martin and Cyton models. The main advantage of Smith-Martin Model lies in the fact that it takes into account the delay in the cell cycle which is not taken into account in models such as the Linear Birth-death ODE Model. However, unless separately determined, the length of Phase B is unknown and there is no information in regular CFSE data on Phase B. This lacking is corrected in the Cyton Model which has advantages as mentioned by Hawkins et al. [14] namely, it is conceptually simple, and flexible for the choice of the distribution of the probability for the events (divide or die). But as we mentioned earlier, it has the disadvantage of assuming that cells behave similarly, and the model becomes computationally expensive.

Given that the mathematical models can be ameliorated over time. We believe that the Smith-Martin Model can be improved; for example, by linking up the duration of A phase and the probability distribution. This will render the model more realistic. Concerning the Cyton Model, the distribution of waiting time to divide and die is chosen to be an exponential function which is not so flexible as the gamma distribution, for instance. In addition, for all models, the term 0th generation is not clear since within a very short time a cell considered as a cell belonging to the 0th generation can divide. Therefore, it is more appropriate to consider the distribution as mentioned by Miao et al. [6] (using the shifted gamma distribution) or by using an empirical distribution which can mirror the behavior of the process of cell division and die.

5. Appendix

5.1 Codes

5.1.1 Smith-Martin model.

The following code helps to select the best parameter to fit the Smith-Martin model to average of the real data from individual-based simulation.

```
1 #####
2 #this code demonstrates the following:
3 #1) defining a function (called smith.martin.model) that simulates the Smith
4   Martin DDE model
5 #2) defining a function (called fit.func) that takes as inputs a parameter
6   vector (called parms)
7 #and a real dataset (called real.data). fit.func passes parms to smith.martin.
8   model,
9 #receives a simulated dataset (called sim.data), and returns the sum of
10  squared differences (SSD) between
11  read.data and sim.data
12 #3) a function that takes as input a real dataset (real.data) and calls an
13  optimising function
14 #(the optim function) to find a parameter vector that minimises the SSD
15  between this dataset and
16  the simulated datasets produced by smith.martin.model
17
18 #####
19
20 #A simple example of how to simulate data from a DDE—i.e. the Smith Martin
21  model
22 #####
23 require(deSolve)
24
25 #this function defines the Smith Martin model
26 smith.martin.model <- function(t,Y,parms){
27   with(as.list(c(parms,Y)), {
28     if(t < deltat){
29       lag1 <- init.vec[1:(ngen+1)]
30     }
31     else{
32       lag1 <- lagvalue(t-deltat)[1:(ngen+1)]
33     }
34     temp <- exp(-d_0*deltat)
35     dN <- rep(0,length(Y))
36     dN[1] <- -(lambda_0 + d_0)*Y[1]
37     dN[2] <- 2*lambda_0*temp*lag1[1] - (lambda+d)*Y[2]
38     dN[3:(ngen+1)] <- 2*lambda*temp*lag1[2:ngen] - (lambda+d)*Y[3:(ngen+1)]
39
40     dN[ngen+2] <- -lambda_0*temp*lag1[1] - d_0*Y[ngen+2]
41     dN[ngen+3] <- 2*lambda_0*temp*lag1[1] - lambda*temp*lag1[2] - d*Y[ngen+3]
```

```

35     dN[(ngen+4):(2*ngen+2)] <- 2*lambda*temp*lag1[2:ngen] - lambda*temp*lag1
36     [3:(ngen+1)] - d*Y[(ngen+4):(2*ngen+2)]
37     list(dN)
38 }
39
40 #this function that takes as inputs a parameter vector (called parms)
41 #and a real dataset (called real.data). It passes parms to smith.martin.model,
42 #receives a simulated dataset (called sim.data), and returns the SSD between
43 #the real and simulated datasets
44 fit.func <- function(parms, real.data){
45     #generated simulated data
46     sim.data <- dede(init.vec, t_pts, smith.martin.model, parms)
47     #put sim.data in the same form as real.data
48     sim.data <- sim.data[, (ngen+3):(2*ngen+3)]
49     #return SSD between sim.data and real.data
50     return (sum((real.data - sim.data)^2))
51 }
52
53 #this function takes as input a real dataset (real.data) and calls an
54 #optimising function
55 #(R's optim function) to find a parameter vector that minimises the SSD
56 #between this dataset and
57 #simulated datasets produced by smith.martin.model
58 optim.func <- function(real.data){
59     #generate new parameter vector
60     lb <- rep(1e-3, 5) #lower-bound of paramter values
61     ub <- c(rep(1, 4), 100) #upper-bound of parameter values
62     parms <- runif(5, min=lb, max=ub) #new random parameter vector
63     names(parms) <- c('lambda_0', 'lambda', 'd_0', 'd', 'deltat') #names of the
64     #parameters
65     opt.parms <- optim(parms, fit.func, gr=NULL, real.data, lower=lb, upper=ub)
66     return (c(opt.parms$value, opt.parms$par))
67 }
68
69 #define parameters of the model
70 #maximum number of division classes to be considered
71 ngen <- 20
72 #vector giving number of cells in each division class
73 init.vec <- rep(0, 2*ngen+2)
74 #number of cells in A phase in i'th generation is denoted NAi
75 #total number of cells in i'th generation is denoted Ni
76 names(init.vec) <- paste(rep(c('NA', 'N'), each=ngen+1), 0:ngen, sep='')
77 #start with 50,000 cells in 0'th generation
78 init.vec[c('NA0', 'N0')] <- 5e4
79 #set the time points at which data should be simulated
80 #these correspond to the time points in the real dataset
81 t_pts <- seq(0, 120, 12)
82 #####
83
84 #read "real" dataset into the computer's memory and the re-shape it
85 #so that column j contains the total number of cells that have divided (j-1)
86     times

```

```

84 #while row i contains cells found at the i'th time point
85 sum.cols <- function(idx, data) return (rowSums(data[,idx]))
86 raw.data <- read.table("~/Dropbox/Data/datasets/agent_based_sim_output_3.txt",
87   header = TRUE)
88 real.data <- matrix(rep(0, (length(t_pts)*(ngen+1))), nrow = length(t_pts))
89 real.data[1,1] <- init.vec[1]
90 real.data[-1,] <- aa <- apply(split(2:43, rep(1:21, each=2)), sum.cols, raw.data)
91 #now, find parameters that minimise the SSD between real.data and
92   corresponding simulated datasets
93 #generated by the Smith Martin model
94 opt.parms <- optim.func(real.data)
95 #the first entry of opt.parms is the SSD while the remaining entries are the
96   corresponding parameter values
97 #A single run of R's optim function typically does not produce the best
98   possible results. It is therefore
99 #advisable call run optim.func many times and choose the parameter values with
100   the lowest SSD.
101 #Below, I call optim.func 10 times and choose the best parameter values
102 opt.parms2 <- mapply(optim.func, rep(list(real.data), 2))
103 best.parms <- opt.parms2[, which.min(opt.parms2[,1])]
104 best.parms

```

Listing 5.1: Smith-Martin selection of the best parameter

And this code fits the Smith-Martin Model with the selected best parameter to the average of real data from individual-based simulation.

```

1 #####
2 #This code fit Smith-Martin model to simulated data
3 #####
4
5 #model definition
6 smith.martin.model <- function(t,Y,parms){
7   with(as.list(c(parms,Y)), {
8     if(t < deltata){
9       lag1 <- init.vec[1:(ngen+1)]
10    }
11    else{
12      lag1 <- lagvalue(t-deltata)[1:(ngen+1)]
13    }
14    temp <- exp(-d_0*deltata)
15    dN <- rep(0, length(Y))
16    dN[1] <- -(lambda_0 + d_0)*Y[1]
17    dN[2] <- 2*lambda_0*temp*lag1[1] - (lambda+d)*Y[2]
18    dN[3:(ngen+1)] <- 2*lambda*temp*lag1[2:ngen] - (lambda+d)*Y[3:(ngen+1)]
19
20    dN[ngen+2] <- -lambda_0*temp*lag1[1] - d_0*Y[ngen+2]
21    dN[ngen+3] <- 2*lambda_0*temp*lag1[1] - lambda*temp*lag1[2] - d*Y[ngen+3]
22    dN[(ngen+4):(2*ngen+2)] <- 2*lambda*temp*lag1[2:ngen] - lambda*temp*lag1
23      [3:(ngen+1)] - d*Y[(ngen+4):(2*ngen+2)]
24    list(dN)
25  })

```

```

25 }
26
27 #this function simulates the Smith–Martin model
28 #and returns the number of cells in each generation
29 #at each time point found in the vector t_pts.
30 simulate.smith.martin.model <- function(parms){
31   out <- dede(init.vec, t_pts, smith.martin.model, parms)
32   return (out[, (ngen+3):(2*ngen+3)])
33 }
34
35 #example run of the mod
36 require(deSolve)
37 #cells in generations ranging from 0 to ngen will be considered
38
39 ngen <- 20
40 init.vec <- rep(0, 2*ngen+2)
41 #number of cells in A phase in i'th generation is denoted NAi
42 #total number of cells in i'th generation is denoted Ni
43 names(init.vec) <- paste(rep(c('NA', 'N'), each=ngen+1), 0:ngen, sep='')
44 #start with 1e6 cells in 0'th generation
45 init.vec[c('NA0', 'N0')] <- 5e4
46 #define parameters of the model
47 parms <- c(lambda_0=1.870167e-02, lambda=3.422586e-02, d_0=1.000000e-03, d
48           =3.425349e-02, deltat=1.000000e-3)
49 #set the time points at which data should be collected
50 t_pts <- seq(0, 120, 12)
51 #run the simulation
52 out <- simulate.smith.martin.model(parms)
53 out
54
55
56
57 sum.cols <- function(idx, data) return (rowSums(data[, idx]))
58
59 #Read the three data set and reshape them to put them in the same shape with
60 #the simulated data#
61 raw.data1 <- read.table("~/Dropbox/Data/datasets/agent_based_sim_output_2.txt",
62                        , header = TRUE)
63 real.data1 <- matrix(rep(0, (length(t_pts)*(ngen+1))), nrow = length(t_pts))
64 real.data1[1,1] <- init.vec[1]
65 real.data1[-1,] <- apply(split(2:43, rep(1:21, each=2)), sum.cols, raw.data1)
66
67 raw.data2 <- read.table("~/Dropbox/Data/datasets/agent_based_sim_output_3.txt",
68                        , header = TRUE)
69 real.data2 <- matrix(rep(0, (length(t_pts)*(ngen+1))), nrow = length(t_pts))
70 real.data2[1,1] <- init.vec[1]
71 real.data2[-1,] <- apply(split(2:43, rep(1:21, each=2)), sum.cols, raw.data2)
72
73 raw.data3 <- read.table("~/Dropbox/Data/datasets/agent_based_sim_output_4.txt",
74                        , header = TRUE)

```

```

74
75 real.data3 <- matrix(rep(0, (length(t_pts)*(ngen+1))), nrow = length(t_pts))
76 real.data3[1,1] <- init.vec[1]
77 real.data3[-1,] <- apply(split(2:43, rep(1:21, each=2)), sum.cols, raw.data3)
78
79 real.data<-(real.data1+ real.data2+ real.data3)/3
80
81 View(real.data)
82
83 ##Plot from gen 0 up to gen 8 the simulate data VS the real data##
84
85 plot(out[,1], xlab = "time points", ylab = "Number of cell Gen 0", type="o", col=
  "red", pch=16)
86 lines(real.data[,1], type="o", col="blue", pch=16)
87 grid(10,10,lwd=1)
88 #legend("topright", legend=c("Real Data", "SM Data", "Cyton Data"), col= c("blue
  ", "red", "cyan"), yjust=1, lty=1, bty="n", cex=1, inset=0)
89
90 plot(out[,2], xlab = "time points", ylab = "Number of cell Gen 1", type="o", col=
  "red", pch=16)
91 lines(real.data[,2], type="o", col="blue", pch=16)
92 grid(10,10,lwd=1)
93 #legend("topright", legend=c("Real Data", "SM Data", "Cyton Data"), col= c("blue
  ", "red", "cyan"), yjust=1, lty=1, bty="n", cex=1, inset=0)
94
95 plot(out[,3], xlab = "time points", ylab = "Number of cell Gen 2", type="o", col=
  "red", pch=16)
96 lines(real.data[,3], type="o", col="blue", pch=16)
97 grid(10,10,lwd=1)
98 #legend("topright", legend=c("Real Data", "SM Data", "Cyton Data"), col= c("blue
  ", "red", "cyan", pch=16), yjust=1, lty=1, bty="n", cex=1, inset=0)
99
100 plot(out[,4], xlab = "time points", ylab = "Number of cell Gen 3", type="o", col=
  "red", pch=16)
101 lines(real.data[,4], type="o", col="blue", pch=16)
102 grid(10,10,lwd=1)
103 #legend("topright", legend=c("Real Data", "SM Data", "Cyton Data"), col= c("blue
  ", "red", "cyan"), yjust=1, lty=1, bty="n", cex=1, inset=0)
104
105 plot(out[,5], xlab = "time points", ylab = "Number of cell Gen 4", type="o", col=
  "red", pch=16)
106 lines(real.data[,5], type="o", col="blue", pch=16)
107 grid(10,10,lwd=1)
108 #legend("topright", legend=c("Real Data", "SM Data", "Cyton Data"), col= c("blue
  ", "red", "cyan"), yjust=1, lty=1, bty="n", cex=1, inset=0)
109
110 plot(out[,6], xlab = "time points", ylab = "Number of cell Gen 5", type="o", col=
  "red", pch=16)
111 lines(real.data[,6], type="o", col="blue", pch=16)
112 grid(10,10,lwd=1)
113 #legend("topright", legend=c("Real Data", "SM Data", "Cyton Data"), col= c("blue
  ", "red", "cyan"), yjust=1, lty=1, bty="n", cex=1, inset=0)
114
115 plot(out[,7], xlab = "time points", ylab = "Number of cell Gen 6", type="o", col=

```

```

      "red", pch=16)
116 lines(real.data[,7], type="o", col="blue", pch=16)
117 grid(10,10,lwd=1)
118 #legend("topright",legend=c("Real Data", "SM Data", "Cyton Data"),col= c("blue
      ", "red", "cyan"),yjust=1,lty=1,bty="n",cex=1, inset=0)
119
120 plot(out[,8],xlab = "time points",ylab = "Number of cell Gen 7",type="o", col=
      "red", pch=16)
121 lines(real.data[,8], type="o", col="blue", pch=16)
122 grid(10,10,lwd=1)
123 #legend("topright",legend=c("Real Data", "SM Data", "Cyton Data"),col= c("blue
      ", "red", "cyan"),yjust=1,lty=1,bty="n",cex=1, inset=0)
124
125
126 plot(out[,9],xlab = "time points",ylab = "Number of cell Gen 8",type="o", col=
      "red", pch=16)
127 lines(real.data[,9], type="o", col="blue", pch=16)
128 grid(10,10,lwd=1)
129 #legend("topright",legend=c("Real Data", "SM Data", "Cyton Data"),col= c("blue
      ", "red", "cyan"),yjust=1,lty=1,bty="n",cex=1, inset=0)

```

Listing 5.2: Fitting Smith-Martin Model to average of real data from individual-based simulation.

5.1.2 Cyton model.

The following code helps to select the best parameter which will be used to fit the Cyton Model to real data from individual-based simulation.

```

1 #####
2 # Select the best parameter for cyton model
3 #####
4 #calculations based on explicit analytic formulas
5 #total number of dividing cell; an explicit expression for  $r_n^{\text{div}}$  (defined
      implicitly
6 #in eqn 13 of Miao et al) was derived and then inserted into eqn 15 of Miao et
      al to get
7 #the total number of cells.
8 sum.rn.div2 <- function(i,a,n,t,theta){
9   j <- 0:i
10   return (((-1)^(n+i+1)/a^(n-i)*(1/theta^(i+1) - exp(-theta*t)*sum(t^(i-j)/
      factorial(i-j)/theta^(j+1))))
11 }
12 sum.rn.div <- function(n,t,par){
13   theta0 <- 1/par['mu0p'] + 1/par['mu0d']
14   if(n == 0){
15     return (par['f0']*N/par['mu0p']/theta0*(1 - exp(-theta0*t)))
16   }
17   else{
18     theta <- 1/par['mup'] + 1/par['mud']
19     a <- theta0 - theta
20     temp <- par['f0']/par['mu0p']*(2*par['f1']/par['mup'])^n*N
21     temp <- temp*(((1/a)^n*(1 - exp(-theta0*t))/theta0) +
      sum(sapply(0:(n-1),sum.rn.div2,a,n,t,theta)))
22   }
23   return (temp)
24 }

```

```

25 }
26 #calculations based on explicit analytic formulas
27 #total number of dieing cells; an explicit expression for  $r_n^{\text{die}}$  (defined
    implicitly
28 #in eqn 14 of Miao et al) was derived and then inserted into eqn 15 of Miao et
    al to get
29 #the total number of cells.
30 sum.rn.die2 <- function(i,a,n,t,theta){
31   j <- 0:(i+1)
32   return ((-1)^(n+i)/a^(n-i-1)*(1/theta^(i+2) - exp(-theta*t)*sum(t^(i-j+1)/
    factorial(i-j+1)/theta^(j+1))))
33 }
34 sum.rn.die <- function(n,t,par){
35   theta0 <- 1/par['mu0p'] + 1/par['mu0d']
36   if(n == 0){
37     return (N/par['mu0d']*((1-par['f0'])*par['mu0d']*(1 - exp(-t/par['mu0d']))
    +
38               par['f0']/theta0*(1 - exp(-theta0*t))))
39   }
40   else{
41     theta <- 1/par['mup'] + 1/par['mud']
42     a <- theta0 - theta; b <- 1/par['mu0d'] - theta
43     temp <- 2*N*par['f0']/par['mu0p']/par['mud']*(2*par['f1']/par['mup'])^(n
    -1)
44     temp2 <- (-1)^(n-1)/a^n*((1 - exp(-theta*t))/theta + (exp(-theta0*t) - 1)/
    theta0) +
45       sum(apply(0:(n-2),sum.rn.die2,a,n,t,theta))*(n > 1)
46     return (temp*temp2)
47   }
48 }
49
50 #this function implements eqn (15) of the Miao et al by calling the other
    functions
51 #defined above
52 sum.cells <- function(n,t,par){
53   return(ifelse(n == 0, N - sum.rn.div(n,t,par) - sum.rn.die(n,t,par),
54               2*sum.rn.div(n-1,t,par) - sum.rn.div(n,t,par) - sum.rn.die(n,t
    ,par)))
55 }
56 temp <- rep(0:ngen,each=length(t_pts))
57 sim.data <- mapply(sum.cells,split(temp,1:length(temp)),split(rep(t_pts,ngen
    +1),1:length(temp)),rep(list(parms),length(temp)))
58 sim.data <- matrix(sim.data,ncol=ngen+1,byrow=FALSE)
59 (sum((sim.data - real.data)^2))
60
61
62 #####
63 # Optimization of smith martin model
64 #####
65 ngen <- 20
66 t_pts <- seq(0,120,12) # initialising default parameters
67 N <- 50000
68 ##### parameter for cyton model #####
69

```



```

70 # read the simulated data generated from agent based model
71 df <- read.table("./datasets/agent_based_sim_output_1.txt", header = TRUE)
72
73 sum.cols <- function(idx, data) return (rowSums(data[,idx]))
74 # reshape data to have total number of cells in each division class in a
    single column
75 real.data <- matrix( rep(0, (length(t_pts)*(ngen+1))), nrow = length(t_pts))
76 real.data[1,1] <- N[1]
77 real.data[-1,] <- apply( split(2:43, rep(1:21, each=2)), sum.cols, df)
78
79 # this function takes as inputs a parameter vector (called parms)
80 #and a "real" dataset (called real.data). It generates a simulated data
81 #(called sim.data) corresponding to input parameters and the calculates
82 #the sum of squared difference (SSD) between real.data and sim.data
83 #generates
84 fit.func <- function(parms, real.data){
85   temp <- rep(0:ngen, each=length(t_pts))
86   sim.data <- mapply(sum.cols, split(temp, 1:length(temp)), split(rep(t_pts, ngen
    +1), 1:length(temp)), rep(list(parms), length(temp)))
87   sim.data <- matrix(sim.data, ncol=ngen+1, byrow=FALSE)
88   return(sum((sim.data - real.data)^2))
89 }
90 #this function takes as input a real dataset (real.data) and calls an
    optimising function
91 #(R's optim function) to find a parameter vector that minimises the SSD
    between this dataset and
92 #simulated datasets produced by the cyton model
93 optim.func <- function(real.data){
94   #generate new parameter vector
95   lb <- rep(1e-3,6) #lower-bound of paramter values
96   ub <- c(rep(1,2), rep(1e3,4)) #upper-bound of parameter values
97   parms <- runif(length(lb), min=lb, max=ub) #new random parameter vector
98   names(parms) <- c('f0', 'f1', 'mu0d', 'mud', 'mu0p', 'mup') #names of the
    parameters
99   opt.parms <- nlminb(parms, fit.func, , , real.data, lower=lb, upper=ub)
100   return (c(opt.parms$objective, opt.parms$par))
101 }
102
103 opt.parms <- mapply(optim.func, rep(list(real.data), 10))
104 best.parms <- opt.parms[, which.min(opt.parms[,1])]

```

Listing 5.3: Selection of the best parameter for Cyton Model

And this code fits the Cyton Model with the selected best parameter to the average of real data from individual-based simulation.

```

1 #####
2 ## This code fit the cyton model to average of simulate data
3 #####
4 ngen <- 20
5 t_pts <- seq(0,120,12)
6 #run the simulation
7
8 #####CYTONMODELS#####

```

```

9 N <- 50000
10 parmsc <- c(f0=1.000000e+00, f1=8.115505e-01, mu0d=4.300041e+02, mud=2.341927e
+01, mu0p=5.316612e+01, mup=1.722716e+01)
11 names(parmsc) <- c('f0', 'f1', 'mu0d', 'mud', 'mu0p', 'mup') #names of the
parameters
12
13 #####
14 # simulate cyton model — modified code by Lamin
15 #####
16 #calculations based on explicit analytic formulas
17 #total number of dividing cell; an explicit expression for  $r_n^{div}$  (defined
implicitly
18 #in eqn 13 of Miao et al) was derived and then inserted into eqn 15 of Miao et
al to get
19 #the total number of cells.
20 sum.rn.div2 <- function(i, a, n, t, theta){
21   j <- 0:i
22   return (((-1)^(n+i+1)/a^(n-i))*(1/theta^(i+1) - exp(-theta*t)*sum(t^(i-j)/
factorial(i-j)/theta^(j+1))))
23 }
24 sum.rn.div <- function(n, t, par){
25   theta0 <- 1/par['mu0p'] + 1/par['mu0d']
26   if(n == 0){
27     return (par['f0']*N/par['mu0p']/theta0*(1 - exp(-theta0*t)))
28   }
29   else{
30     theta <- 1/par['mup'] + 1/par['mud']
31     a <- theta0 - theta
32     temp <- par['f0']/par['mu0p']*(2*par['f1']/par['mup'])^n*N
33     temp <- temp*(((1/a)^n*(1 - exp(-theta0*t))/theta0) +
sum(apply(0:(n-1), sum.rn.div2, a, n, t, theta)))
34     return (temp)
35   }
36 }
37 }
38 #calculations based on explicit analytic formulas
39 #total number of dieing cells; an explicit expression for  $r_n^{die}$  (defined
implicitly
40 #in eqn 14 of Miao et al) was derived and then inserted into eqn 15 of Miao et
al to get
41 #the total number of cells.
42 sum.rn.die2 <- function(i, a, n, t, theta){
43   j <- 0:(i+1)
44   return (((-1)^(n+i)/a^(n-i-1))*(1/theta^(i+2) - exp(-theta*t)*sum(t^(i-j+1)/
factorial(i-j+1)/theta^(j+1))))
45 }
46 sum.rn.die <- function(n, t, par){
47   theta0 <- 1/par['mu0p'] + 1/par['mu0d']
48   if(n == 0){
49     return (N/par['mu0d']*((1-par['f0'])*par['mu0d']*(1 - exp(-t/par['mu0d']))
+
par['f0']/theta0*(1 - exp(-theta0*t))))
50   }
51   else{
52     theta <- 1/par['mup'] + 1/par['mud']
53

```

```

54   a <- theta0 - theta; b <- 1/par['mu0d'] - theta
55   temp <- 2*N*par['f0']/par['mu0p']/par['mud']*(2*par['f1']/par['mup'])^(n
56   -1)
57   temp2 <- (-1)^(n-1)/a^n*((1 - exp(-theta*t))/theta + (exp(-theta0*t) - 1)/
58   theta0) +
59   sum(sapply(0:(n-2),sum.rn.die2,a,n,t,theta))*(n > 1)
60   return (temp*temp2)
61 }
62 }
63 #this function implements eqn (15) of the Miao et al by calling the other
64 #functions
65 #defined above
66 sum.cells <- function(n,t,par){
67   return(ifelse(n == 0, N - sum.rn.div(n,t,par) - sum.rn.die(n,t,par),
68   2*sum.rn.div(n-1,t,par) - sum.rn.div(n,t,par) - sum.rn.die(n,t
69   ,par)))
70 }
71 temp <- rep(0:ngen,each=length(t_pts))
72 sim.data <- mapply(sum.cells,split(temp,1:length(temp)),split(rep(t_pts,ngen
73 +1),1:length(temp)),rep(list(parmsc),length(temp)))
74 sim.data <- matrix(sim.data,ncol=ngen+1,byrow=FALSE)
75 #####Fitting the models to data#####
76
77 sum.cols <- function(idx, data) return (rowSums(data[,idx]))
78
79 #Read the three data set and reshape them to put them in the same shape with
80 the simulated data#
81 raw.data1 <- read.table("~/Dropbox/Data/datasets/agent_based_sim_output_2.txt"
82 ,header = TRUE)
83 real.data1 <- matrix(rep(0, (length(t_pts)*(ngen+1))), nrow = length(t_pts))
84 real.data1[1,1] <- init.vec[1]
85 real.data1[-1,] <- sapply(split(2:43,rep(1:21,each=2)),sum.cols,raw.data1)
86
87 raw.data2 <- read.table("~/Dropbox/Data/datasets/agent_based_sim_output_3.txt"
88 ,header = TRUE)
89 real.data2 <- matrix(rep(0, (length(t_pts)*(ngen+1))), nrow = length(t_pts))
90 real.data2[1,1] <- init.vec[1]
91 real.data2[-1,] <- sapply(split(2:43,rep(1:21,each=2)),sum.cols,raw.data2)
92
93 raw.data3 <- read.table("~/Dropbox/Data/datasets/agent_based_sim_output_4.txt"
94 ,header = TRUE)
95 real.data3 <- matrix(rep(0, (length(t_pts)*(ngen+1))), nrow = length(t_pts))
96 real.data3[1,1] <- init.vec[1]
97 real.data3[-1,] <- sapply(split(2:43,rep(1:21,each=2)),sum.cols,raw.data3)
98

```

```

99 #####Plot from generation 0 up to generation 8 the simulate data VS the real
    data#####
100
101 plot(sim.data[,1],xlab = "time points",ylab = "Number of cell Gen 0",type="o"
    , col="magenta",pch=16)
102 lines(real.data[,1], type="o", col="blue",pch=16)
103 grid(10,10,lwd=1)
104 #legend("topright",legend=c("Real Data", "SM Data", "Cyton Data"),col= c("blue
    ","red","cyan"),yjust=1,lty=1,bty="n",cex=1, inset=0)
105
106 plot(sim.data[,2],xlab = "time points",ylab = "Number of cell Gen 1",type="o",
    col="magenta",pch=16)
107 lines(real.data[,2], type="o", col="blue",pch=16)
108 grid(10,10,lwd=1)
109 #legend("topright",legend=c("Real Data", "SM Data", "Cyton Data"),col= c("blue
    ","red","cyan"),yjust=1,lty=1,bty="n",cex=1, inset=0)
110
111 plot(real.data[,3],xlab = "time points",ylab = "Number of cell Gen 2",type="o"
    , col="blue",pch=16)
112 lines(sim.data[,3], type="o", col="magenta",pch=16)
113 grid(10,10,lwd=1)
114 #legend("topright",legend=c("Real Data", "Cyton Data"),col= c("blue","magenta
    "),yjust=1,lty=1,bty="n",cex=1, inset=0)
115
116 plot(real.data[,4],xlab = "time points",ylab = "Number of cell Gen 3",type="o"
    , col="blue",pch=16)
117 lines(sim.data[,4], type="o", col="magenta",pch=16)
118 grid(10,10,lwd=1)
119 #legend("topright",legend=c("Real Data", "Cyton Data"),col= c("blue","magenta
    "),yjust=1,lty=1,bty="n",cex=1, inset=0)
120
121 plot(real.data[,5],xlab = "time points",ylab = "Number of cell Gen 4",type="o"
    , col="blue",pch=16)
122 lines(sim.data[,5], type="o", col="magenta",pch=16)
123 grid(10,10,lwd=1)
124 #legend("topright",legend=c("Real Data", "Cyton Data"),col= c("blue","magenta
    "),yjust=1,lty=1,bty="n",cex=1, inset=0)
125
126 plot(real.data[,6],xlab = "time points",ylab = "Number of cell Gen 5",type="o"
    , col="blue",pch=16)
127 lines(sim.data[,6], type="o", col="magenta",pch=16)
128 grid(10,10,lwd=1)
129 #legend("topright",legend=c("Real Data", "Cyton Data"),col= c("blue","magenta
    "),yjust=1,lty=1,bty="n",cex=1, inset=0)
130
131 plot(real.data[,7],xlab = "time points",ylab = "Number of cell Gen 6",type="o"
    , col="blue",pch=16)
132 lines(sim.data[,7], type="o", col="magenta",pch=16)
133 grid(10,10,lwd=1)
134 #legend("topright",legend=c("Real Data", "Cyton Data"),col= c("blue","magenta
    "),yjust=1,lty=1,bty="n",cex=1, inset=0)
135
136 plot(real.data[,8],xlab = "time points",ylab = "Number of cell Gen 7",type="o"
    , col="blue",pch=16)

```

```
137 lines(sim.data[,8], type="o", col="magenta",pch=16)
138 grid(10,10,lwd=1)
139 #legend("topright",legend=c("Real Data", "Cyton Data"),col= c("blue","magenta")
140     ),yjust=1,lty=1,bty="n",cex=1, inset=0)
141 plot(real.data[,9],xlab = "time points",ylab = "Number of cell Gen 8",type="o"
142     , col="blue",pch=16)
143 lines(sim.data[,9], type="o", col="magenta",pch=16)
144 grid(10,10,lwd=1)
145 #legend("topright",legend=c("Real Data", "Cyton Data"),col= c("blue","magenta")
146     ),yjust=1,lty=1,bty="n",cex=1, inset=0)
```

Listing 5.4: Fitting Cyton Model to average of real data

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